TRIANGLES IN THE PLANE: A SELF-CONTAINED PROOF BLUEPRINT

Abstract. This blueprint expands the argument of the supplied manuscript on planar triangles. It includes proofs of the three-letter density Hales–Jewett theorem and the analytic representation and rigidity facts needed by the argument. Weighted circle compactness, measurable stationarity, annular cancellation, and frequency summation then give uniform smoothing of the triangle counting form and the geometric conclusion. All results are arranged in dependency order, and each theorem, proposition, lemma, and corollary after the introduction is followed immediately by its proof. The foundational measure theory and calculus assumed are specified below.

Contents

1. Introduction and foundations 1
Foundations and the scope of self-containment 2
2. Analytic prerequisites with proofs 2
2.1. Fourier analysis in the required form 3
2.2. The compactness facts used for circle operators 6
2.3. Spectral measures of the correlation functions 7
2.4. Local matrix groups and analytic zero sets 11
3. Finite words, binary density, and tilings 12
3.1. Equal-slices averaging and the three-letter theorem 16
4. Geometry, forms, and elementary estimates 21
5. Finite-dimensional scalar kernels 25
6. Analytic pieces and measurable rigidity 27
7. Translation correlations and local cancellation 30
8. Localization and a global annular estimate 36
9. Summation over frequency annuli 37
10. Approximation of the counting form 41
11. From smoothing to triangles 42
References 45

1. Introduction and foundations

The purpose of this document is to make the proof structure and the measure-theoretic, functional-analytic, and summation steps of the supplied paper explicit. The notation is kept close to that paper. Definitions are made before use, and the geometric density argument is placed after the analytic estimates on which it depends.

For a measurable set $E \subset \mathbb{R}^2$, write

$$d^*(E) = \limsup_{L\to\infty} \sup_{z\in\mathbb{R}^2} \frac{|E \cap (z + [0, L]^2)|}{L^2}.$$

Here and throughout, $|\cdot|$ denotes Lebesgue measure when applied to a measurable subset of Euclidean space.

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Theorem A (Geometric conclusion). Let $\Delta \subset \mathbb{R}^2$ consist of three noncollinear points. If $E \subset \mathbb{R}^2$ is measurable and $d^*(E) > 0$, there is $\lambda_0 > 0$ such that, for every $\lambda \geq \lambda_0$, the set E contains a translate of a rotation of $\lambda\Delta$.

The analytic assertion used to prove this is as follows. Let $J(x_1, x_2) = (-x_2, x_1)$ and $M = \alpha I + \beta J$ with $\beta \neq 0$. Let σ be uniform probability measure on the unit circle. For a finite measure ν define

$$\Lambda_\nu(f_0, f_1, f_2) = \iint f_0(x) f_1(x+u) f_2(x+Mu)\, \mathrm{d}\nu(u)\, \mathrm{d}x.$$

Write ν_s for the pushforward of ν under $u \mapsto su$, and write $\|\Lambda_\nu\|$ for the $L^2 \times L^2 \times L^\infty$ operator norm.

Theorem B (Uniform smoothing). For every nonnegative $\rho \in L^1(\mathbb{R}^2)$ with $\int \rho = 1$, put $\rho_\varepsilon(u) = \varepsilon^{-2}\rho(u/\varepsilon)$. Then

$$\limsup_{\varepsilon \downarrow 0} \sup_{s>0} \|\Lambda_{(\sigma * \rho_\varepsilon - \sigma)_s}\| = 0.$$

The same statement holds with the L^∞ slot at any of the three vertices.

These two introductory statements are proved in the main text, after their prerequisites. The analytic proof has three main parts: a rigidity theorem for measurable stationary frequency fields; a compactness argument for cancellation on a single frequency annulus; and a positive telescoping estimate that sums the annuli without a decay rate.

Foundations and the scope of self-containment. We work over the usual construction of Lebesgue measure, integration, L^p spaces, and Hilbert spaces. The foundational facts used are Tonelli and Fubini, monotone and dominated convergence, regularity of finite Borel measures on Euclidean spaces, completion of Lebesgue measure, measure extension, Radon–Nikodym, and monotone classes. We also use elementary Euclidean calculus, including change of variables, the inverse function theorem, and convergent power series. The Hilbert-space foundations include orthogonal projection onto a closed subspace and the existence of adjoints of bounded operators. Basic inner-product algebra and finite combinatorics are understood. Thus "self-contained" here means a proof over these standard analysis foundations, rather than a reconstruction of measure theory itself.

All additional inputs used by the triangle argument are proved in the text. The analytic preliminaries include the needed Fourier facts, sequential weak-* extraction, elementary compact-operator facts, the positive-definite representation theorem, the exponential form of local matrix groups, and the real-analytic zero-set argument. These statements are restricted to the settings in which they are applied.

The combinatorial preliminaries prove the density Hales–Jewett theorem for three letters. This is the theorem of Furstenberg and Katznelson [1]; the proof here follows the finite density-increment approach of Polymath [2], with a tiling argument in the form developed by Dodos, Kanellopoulos, and Tyros [3]. The two-letter input, the comparisons of finite probability measures, the insensitive-set tilings, and the density-increment iteration are all proved. The three-letter theorem is therefore an established earlier result when it is used in the final geometric argument.

2. Analytic prerequisites with proofs

This section supplies the Fourier, compactness, spectral-measure, matrix-group, and analytic-zero-set facts used in the argument. The foundations are Lebesgue measure and integration, the ordinary extension theorem for a finite premeasure, elementary Euclidean calculus, and the elementary Hilbert-space operations of taking adjoints and orthogonal projections. No spectral theorem for compact operators, representation theorem for positive-definite functions, or local Lie-group theorem is assumed.

2.1. **Fourier analysis in the required form.**

Definition 2.1 (Fourier convention and Gaussian kernels)**.** For $f \in L^1(\mathbb{R}^2)$ and a finite complex Borel measure η, put

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x) \exp(-2\pi i x \cdot \xi)\, \mathrm{d}x, \qquad \widehat{\eta}(\xi) = \int_{\mathbb{R}^2} \exp(-2\pi i x \cdot \xi)\, \mathrm{d}\eta(x).$$

The inverse transform uses the positive sign in the exponential and is denoted by a check. The Schwartz space $\mathcal{S}(\mathbb{R}^2)$ consists of the smooth functions for which every derivative, multiplied by any polynomial, is bounded. For $t > 0$ write

$$\gamma_t(x) = t^{-1} \exp(-\pi |x|^2/t).$$

Lemma 2.2 (The convolution inequalities used below)**.** *If κ is a finite complex Borel measure and $p \in \{1, 2, \infty\}$, then*

$$\|\kappa * f\|_p \le \|\kappa\|_{\mathrm{TV}} \|f\|_p.$$

In particular this holds when κ has an L^1 density.

Proof. The assertion for $p = \infty$ follows by taking absolute values inside the convolution integral; for $p = 1$ it follows by Tonelli and translation invariance. For $p = 2$, Cauchy–Schwarz against the finite measure $|\kappa|$ gives

$$|(\kappa * f)(x)|^2 \le \|\kappa\|_{\mathrm{TV}} \int |f(x - y)|^2 \, \mathrm{d}|\kappa|(y).$$

Tonelli and translation invariance give the claimed L^2 bound after integration in x. The same estimate shows that the defining integral is absolutely convergent for almost every x, by another application of Cauchy–Schwarz. □

Lemma 2.3 (Density and continuity under translations and dilations)**.** *Smooth compactly supported functions are dense in $L^p(\mathbb{R}^2)$ for $p = 1, 2$. For these two exponents, translations are continuous:*

$$\|f(\cdot - h) - f\|_p \longrightarrow 0 \qquad (h \to 0).$$

For $F \in L^p(\mathbb{R}^2 \times [0, 2\pi))$, the same conclusion holds for translation in the first two variables, with the product measure $\mathrm{d}x\, \mathrm{d}\theta/(2\pi)$. The normalized dilations $D_r f(x) = r^{-2} f(x/r)$ satisfy $\|D_r f - f\|_1 \to 0$ as $r \to 1$.

Proof. Truncating both the support and the magnitude of a function reduces density to bounded functions on a bounded set. Simple functions approximate these in L^p. By regularity of Lebesgue measure, their level sets can be approximated in measure by finite unions of bounded rectangles. The indicator of a bounded rectangle can in turn be approximated in L^p by a smooth function between zero and one that equals one away from a thin boundary layer and vanishes outside a slightly enlarged rectangle. These steps prove the density assertion.

For a smooth compactly supported function, translation continuity follows from uniform continuity and a common bounded support for small translations. Translation is an isometry on L^p, so approximation proves the assertion for every $f \in L^p$. The same argument proves dilation continuity in L^1, using $\|D_r f\|_1 = \|f\|_1$ and the common bounded support of $D_r g$ for r near one and $g \in C_c^\infty$. For the product-space assertion, apply planar translation continuity to $F(\cdot, \theta)$ for almost every θ. Fubini writes the pth power of the product norm as the integral of the pth powers of these sectional norms. They are dominated by $2^p \|F(\cdot, \theta)\|_p^p$, an integrable function of θ. Dominated convergence gives the assertion. □

Lemma 2.4 (The Gaussian Fourier calculation)**.** *The Gaussian kernels have integral one and satisfy*

$$\widehat{\gamma_t}(\xi) = \exp(-\pi t |\xi|^2).$$

If f is bounded and continuous, then $\gamma_t * f(x) \to f(x)$ at every x as $t \downarrow 0$. The convergence is uniform when f is bounded and uniformly continuous.

Proof. The elementary Gaussian integral is $\int_{\mathbb{R}} \exp(-\pi x^2)\,\mathrm{d}x = 1$: square the integral and use polar coordinates in the plane. If

$$G(\xi) = \int_{\mathbb{R}} \exp(-\pi x^2)\exp(-2\pi i x\xi)\,\mathrm{d}x,$$

differentiation under the integral and integration by parts give $G'(\xi) = -2\pi\xi G(\xi)$. Since $G(0) = 1$, differentiating $\exp(\pi\xi^2)G(\xi)$ shows that $G(\xi) = \exp(-\pi\xi^2)$. Taking products of the one-dimensional identity and then scaling proves the displayed formula and the integral normalization.

After the substitution $y = \sqrt{t}\,z$,

$$(\gamma_t * f)(x) - f(x) = \int \exp(-\pi|z|^2)(f(x - \sqrt{t}\,z) - f(x))\,\mathrm{d}z.$$

Pointwise convergence follows by dominated convergence. For uniform convergence, use uniform continuity on $|z| \le A$ and bound the remaining integral by $2\|f\|_\infty \int_{|z|>A} \exp(-\pi|z|^2)\,\mathrm{d}z$; first increase A, and then decrease t. □

Lemma 2.5 (Fourier inversion and Plancherel)**.** *The Fourier transform maps $\mathcal{S}(\mathbb{R}^2)$ onto itself, and*

$$f(x) = \int \widehat{f}(\xi)\exp(2\pi i x\cdot\xi)\,\mathrm{d}\xi \qquad (f \in \mathcal{S}).$$

For $f, g \in \mathcal{S}$,

$$\int \widehat{f}(\xi)\overline{\widehat{g}(\xi)}\,\mathrm{d}\xi = \int f(x)\overline{g(x)}\,\mathrm{d}x.$$

Consequently the Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^2)$.

Proof. Differentiation under the integral expresses derivatives of $\widehat{f}$ as transforms of polynomial multiples of f. Integration by parts expresses polynomial multiples of these derivatives as transforms of derivatives of polynomial multiples of f. All the latter functions are integrable. This proves that $\widehat{f}$ is Schwartz. The identical argument applies to the inverse transform.

Fubini and Lemma 2.4 give, for $t > 0$,

$$\int \widehat{f}(\xi)\exp(2\pi i x\cdot\xi)\exp(-\pi t|\xi|^2)\,\mathrm{d}\xi = (f * \gamma_t)(x).$$

The left side converges to the inverse-transform integral by dominated convergence, because $\widehat{f} \in L^1$. The right side converges to $f(x)$ by Lemma 2.4. This proves inversion, including that the transform is onto the Schwartz space.

Apply inversion to g, take its complex conjugate, multiply by f, and integrate. Fubini is legitimate because $\|f\|_1\|\widehat{g}\|_1 < \infty$. The result is the inner-product identity. In particular the transform is an L^2 isometry on the Schwartz space. This space is dense in L^2 by Lemma 2.3, so the isometry extends uniquely to L^2. Its range is closed and contains the dense Schwartz space, and is therefore all of L^2. □

Lemma 2.6 (Uniqueness for finite measures)**.** *A finite complex Borel measure on $\mathbb{R}^2$ is determined by its Fourier transform. In particular, if ν is a finite positive measure, the characters $\xi \mapsto \exp(2\pi i s\cdot\xi)$ with $s \in \mathbb{Q}^2$ span a dense subspace of $L^2(\nu)$.*

Proof. It is enough to consider a finite complex measure η with $\widehat{\eta} = 0$. The Gaussian formula and Fubini imply

$$(\eta * \gamma_t)(x) = \int \exp(2\pi i x\cdot\xi)\exp(-\pi t|\xi|^2)\widehat{\eta}(\xi)\,\mathrm{d}\xi = 0.$$

For $\varphi \in C_c(\mathbb{R}^2)$, another use of Fubini gives

$$0 = \int \varphi(x)(\eta * \gamma_t)(x)\,\mathrm{d}x = \int (\varphi * \gamma_t)(y)\,\mathrm{d}\eta(y).$$

By uniform convergence in Lemma 2.4, the last integral tends to $\int \varphi \, d\eta$. Thus this integral is zero for every continuous compactly supported φ. Regularity of a finite Borel measure on Euclidean space shows that these test functions determine the measure: approximate indicators of compact sets from above by continuous cutoffs in arbitrarily small open neighborhoods, and then use inner regularity on Borel sets. Hence $\eta = 0$.

If $g \in L^2(\nu)$ is orthogonal to all the stated characters, the finite complex measure $\overline{g}\,\nu$ has Fourier transform zero on $\mathbb{Q}^2$, with the sign of the argument immaterial. Its transform is continuous by dominated convergence, so it is zero everywhere. The first assertion therefore gives $g = 0$ almost everywhere. The orthogonal complement of the character span is zero, proving density. □

Definition 2.7 (Fourier support for bounded functions)**.** For a compact set $K \subset \mathbb{R}^2$, a bounded measurable function g has Fourier support in K if

$$\int_{\mathbb{R}^2} g(x)\widehat{\varphi}(x)\,dx = 0 \quad \text{for every } \varphi \in C_c^\infty(\mathbb{R}^2) \text{ with } \operatorname{supp}\varphi \cap K = \varnothing.$$

This is the test-function definition of distributional Fourier support; the displayed integrals are absolutely convergent because $\widehat{\varphi}$ is Schwartz. For an L^2 function, Fourier support in K means that its L^2 Fourier transform vanishes almost everywhere outside K.

Lemma 2.8 (Reproduction of an input with compact Fourier support)**.** *Suppose $p \in \mathcal{S}(\mathbb{R}^2)$ and $\widehat{p} = 1$ on a neighborhood of a compact set K. If $g \in L^2$ or $g \in L^\infty$ has Fourier support in K, then $p * g = g$ almost everywhere. For a function in $L^2 \cap L^\infty$, the two support conditions in Definition 2.7 agree.*

Proof. For L^2 inputs, the identity $\widehat{p * g} = \widehat{p}\,\widehat{g}$ holds first for Schwartz functions by Fubini and then by L^2 approximation, using Lemmas 2.2 and 2.5. The support assumption then proves the assertion by injectivity of the L^2 Fourier transform.

For bounded inputs we first extend the defining support condition to Schwartz tests φ that vanish on a neighborhood of K. Choose $\eta \in C_c^\infty$ equal to one near zero and put $\varphi_R(\xi) = \eta(\xi/R)\varphi(\xi)$. Each φ_R is an allowed compactly supported test. Write $\Delta = \partial_1^2 + \partial_2^2$ for the Laplacian. If $v \in \mathcal{S}$, integration by parts gives

$$(1 + 4\pi^2|x|^2)^2\widehat{v}(x) = \widehat{(1-\Delta)^2 v}(x), \qquad \|\widehat{v}\|_1 \le C \sum_{|\alpha|\le 4} \|\partial^\alpha v\|_1.$$

The second inequality follows by bounding the transform on the right by the L^1 norm of its input and integrating $(1 + 4\pi^2|x|^2)^{-2}$ over the plane. The product rule and rapid decay of φ give $\|\partial^\alpha(\varphi_R - \varphi)\|_1 \to 0$ for every $|\alpha| \le 4$: the undifferentiated cutoff contributes a tail integral, and each differentiated cutoff contributes a factor R^{-1} or smaller times a rapidly decaying derivative. Consequently $\|\widehat{\varphi}_R - \widehat{\varphi}\|_1 \to 0$. Pairing with bounded g proves $\int g\widehat{\varphi} = 0$.

For any $\psi \in \mathcal{S}$, Fubini now gives

$$\int (p * g - g)(x)\widehat{\psi}(x)\,dx = \int g(y)\widehat{(p-1)\psi}(y)\,dy = 0.$$

For the convolution term, absolute integrability follows from $\|g\|_\infty \|p\|_1 \|\widehat{\psi}\|_1 < \infty$. The final equality holds because $(\widehat{p} - 1)\psi$ is Schwartz and vanishes near K. Lemma 2.5 says that Fourier transforms of Schwartz tests exhaust the Schwartz space. Thus the bounded function $p * g - g$ annihilates every C_c^∞ test and, by their L^1 density from Lemma 2.3, every L^1 test. It is zero almost everywhere.

Finally, for $g \in L^2 \cap L^\infty$, Plancherel gives $\int g\widehat{\varphi} = \int \widehat{g}\,\varphi$. Vanishing of the latter pairings for all smooth tests supported outside K is equivalent to $\widehat{g} = 0$ there almost everywhere. For the nontrivial direction, cover $\mathbb{R}^2 \setminus K$ by countably many balls whose closures are disjoint from K. Smooth tests supported in a ball are dense in its L^2 space: first multiply a function by smooth cutoffs increasing to one inside that ball, and then approximate each resulting function in $L^2(\mathbb{R}^2)$ using Lemma 2.3, multiplying the approximants by another cutoff equal to one on its compact

support and supported in the ball. Thus $\widehat{g}$ vanishes on each ball almost everywhere. This proves agreement of the definitions. □

2.2. **The compactness facts used for circle operators.**

Definition 2.9 (Compact operator)**.** A bounded linear operator between Hilbert spaces is compact if the image of every bounded sequence has a norm-convergent subsequence. Equivalently, the image of its unit ball has a finite ε-net for every $\varepsilon > 0$.

Lemma 2.10 (Elementary compactness principles)**.** *For a bounded operator between Hilbert spaces, $\|T\| = \|T^*\|$ and $\|TT^*\| = \|T\|^2$. The following compactness facts also hold.*

(1) Finite-rank operators are compact, and an operator-norm limit of compact operators is compact.

(2) An integral operator with kernel in $L^2(X \times Y)$, where X, Y are bounded measurable subsets of Euclidean space, is compact from $L^2(Y)$ to $L^2(X)$.

(3) If T is compact and a bounded sequence f_n converges weakly to f, then $Tf_n \to Tf$ in norm.

Proof. Taking the supremum of $|\langle Tx, y\rangle|$ over unit vectors in the two spaces gives $\|T\| = \|T^*\|$. The inequality $\|TT^*\| \le \|T\|^2$ follows by composition, and $\|T^*y\|^2 = \langle TT^*y, y\rangle$ gives the reverse inequality after taking the supremum over unit y.

The first assertion for finite rank follows from compactness of bounded closed sets in finite-dimensional Euclidean space. If $T_m \to T$ in operator norm, approximate the image of the unit ball under T by its image under one T_m and then by a finite net for that image. This gives a finite net for T at every positive accuracy. A successive-net or diagonal argument shows that every sequence in this image has a Cauchy subsequence; completeness makes it convergent.

For the second assertion, finite sums of kernels $a(x)b(y)$ are dense in $L^2(X\times Y)$. Indeed, rectangular simple functions are dense in the product measure space, by approximation of measurable sets using the algebra of finite unions of measurable rectangles. Such finite sums define finite-rank operators. Cauchy–Schwarz shows that the operator norm of the difference of two integral operators is at most the L^2 norm of the difference of their kernels. Apply the first assertion.

For the third, suppose norm convergence fails. A subsequence stays a positive distance from Tf, and a further subsequence converges in norm to some v by compactness. For every test vector h, the identity $\langle Tf_n, h\rangle = \langle f_n, T^*h\rangle$ shows that the weak limit is Tf. The norm limit v must therefore equal Tf, a contradiction. □

Lemma 2.11 (Compactness from the product with the adjoint)**.** *If $T : H \to H'$ is bounded and $TT^* : H' \to H'$ is compact, then T is compact.*

Proof. For a bounded sequence u_n in H', choose a subsequence on which TT^*u_n converges. On that subsequence,

$$\|T^*(u_n - u_m)\|^2 = \langle TT^*(u_n - u_m), u_n - u_m\rangle \longrightarrow 0.$$

Thus T^* is compact. Given $\varepsilon > 0$, choose a finite ε-net for T^* of the unit ball. Let $F \subset H$ be the span of its finitely many centers and P_F the orthogonal projection onto F. Then $\|(I - P_F)T^*\| \le \varepsilon$. Taking adjoints gives $\|T(I - P_F)\| \le \varepsilon$. The operator TP_F has finite rank and approximates T in norm. Lemma 2.10 proves compactness. □

Lemma 2.12 (A gap for a nonnegative compact perturbation)**.** *Let K be a compact self-adjoint operator on a Hilbert space, and suppose $B = I + K$ is nonnegative. Then* $\ker B$ *is finite dimensional, and there is $\gamma > 0$ such that*

$$\langle Bf, f\rangle \ge \gamma \operatorname{dist}(f, \ker B)^2 \qquad (f \in H).$$

Proof. If the kernel were infinite dimensional, it would contain an orthonormal sequence u_n. Since $Ku_n = -u_n$, its image under K would have no convergent subsequence, contrary to compactness.

For any bounded nonnegative operator B, positivity of $\langle B(x+ty), x+ty\rangle$ for real or complex t gives the Cauchy–Schwarz inequality for its quadratic form:

$$|\langle Bx, y\rangle|^2 \le \langle Bx, x\rangle\langle By, y\rangle.$$

When the second diagonal term is positive, minimize the quadratic polynomial in t; when it is zero, letting t vary first shows that the cross term is zero. Taking the supremum over unit vectors y gives

$$\|Bx\|^2 \le \|B\|\langle Bx, x\rangle.$$

If there were no positive lower bound for the quadratic form on unit vectors orthogonal to $\ker B$, there would be such vectors x_n with $\langle Bx_n, x_n\rangle \to 0$. The last inequality gives $Bx_n \to 0$. On a subsequence Kx_n converges, so $x_n = Bx_n - Kx_n$ converges in norm to a unit vector x. This vector is still orthogonal to the kernel, but $Bx = 0$, a contradiction. Finally decompose an arbitrary f into its kernel component and its orthogonal component. The quadratic form vanishes on the first component and has no cross term, proving the stated estimate. □

Lemma 2.13 (Sequential weak-* extraction)**.** *On $\mathbb{R}^2$, and on $\mathbb{R}^2 \times [0, 2\pi)$ with measure $\mathrm{d}x\,\mathrm{d}\theta/(2\pi)$, every sequence bounded in L^∞ has a weak-* convergent subsequence. A common subsequence can be chosen for any countable family of such bounded sequences.*

Proof. Write the measure as m. Step functions on rectangles with rational endpoints and rational real and imaginary values form a countable dense subset of L^1, after allowing finite linear combinations. Density follows by truncation and regularity of Lebesgue measure, as in Lemma 2.3. If $\|f_n\|_\infty \le A$, the pairings of f_n with each of these tests form a bounded scalar sequence. Successive subsequence extraction and a diagonal choice make all these pairings converge. The bound $A\|\varphi\|_1$ then extends the limit to a bounded linear functional L on all of L^1, and gives convergence against every L^1 test.

For clarity, this functional is represented by a bounded function without invoking a compactness theorem for dual spaces. Exhaust the underlying space by sets X_k of finite measure. On each X_k, the set function $A \mapsto L(\mathbf{1}_A)$ is a complex measure: for disjoint sets, countable additivity follows from convergence of the partial union indicators in $L^1(X_k)$. Its total variation is at most Am, by the defining norm bound. The Radon–Nikodym theorem gives a density g_k with $|g_k| \le A$. Uniqueness of densities makes these functions agree on overlaps. They define a global $g \in L^\infty$ satisfying $L(\varphi) = \int g\varphi\,\mathrm{d}m$, first for simple functions of finite support and then for all L^1 functions by density. This is the required weak-* limit.

For countably many sequences, perform the scalar diagonal selection over the countable set of all pairs consisting of a sequence index and a test index. The preceding construction then supplies the limit for each sequence on the same subsequence. □

2.3. **Spectral measures of the correlation functions.**

Definition 2.14 (Positive-definite function)**.** A function $c : \mathbb{R}^2 \to \mathbb{C}$ is positive definite if

$$\sum_{a,b=1}^{N} z_a\overline{z_b}\,c(s_a - s_b) \ge 0$$

for every finite family $s_a \in \mathbb{R}^2$, $z_a \in \mathbb{C}$; the displayed quantity is required to be real.

Lemma 2.15 (Integrable positive-definite functions)**.** *If c is positive definite, then $c(0) \ge 0$, $c(-s) = \overline{c(s)}$, and $|c(s)| \le c(0)$. If, in addition, c is continuous and integrable, then $\widehat{c}$ is a nonnegative integrable function,*

$$\int \widehat{c}(\xi)\,\mathrm{d}\xi = c(0), \qquad c(s) = \int \exp(2\pi i s\cdot\xi)\widehat{c}(\xi)\,\mathrm{d}\xi.$$

Proof. Use one point and then the two points $0, s$ in the defining inequality. Varying their complex coefficients shows that the corresponding matrix is Hermitian and nonnegative. Minimizing its quadratic form gives $|c(s)|^2 \le c(0)^2$. This also shows that c is identically zero if $c(0) = 0$.

For a continuous compactly supported function φ, the defining finite inequalities and Riemann-sum approximation give

$$\iint c(x-y)\varphi(x)\overline{\varphi(y)}\,\mathrm{d}x\,\mathrm{d}y \ge 0.$$

Fix $\xi \in \mathbb{R}^2$ and a real $\varphi \in C_c^\infty$ with $\|\varphi\|_2 = 1$. Apply this inequality to $L^{-1}\varphi(x/L)\exp(-2\pi i\xi\cdot x)$. After writing $h = x - y$, the resulting integral is

$$\int c(h)\exp(-2\pi i\xi\cdot h)\left(\int \varphi(t+h/L)\varphi(t)\,\mathrm{d}t\right)\mathrm{d}h.$$

The inner correlation has absolute value at most one and tends to one for each fixed h, by translation continuity in L^2. Since $c \in L^1$, dominated convergence shows that the last integral tends to $\widehat{c}(\xi)$. Thus $\widehat{c}$ is nonnegative everywhere.

For $t > 0$, Fubini and Lemma 2.4 give

$$\int \widehat{c}(\xi)\exp(-\pi t|\xi|^2)\,\mathrm{d}\xi = \int c(s)\gamma_t(s)\,\mathrm{d}s \longrightarrow c(0).$$

Monotone convergence on the left as $t \downarrow 0$ proves integrability and the mass formula. Finally,

$$\int \widehat{c}(\xi)\exp(2\pi i s\cdot\xi)\exp(-\pi t|\xi|^2)\,\mathrm{d}\xi = (c * \gamma_t)(s).$$

The left side converges by the newly proved integrability, and the right side converges by continuity and boundedness of c. This proves the representation. □

Lemma 2.16 (A subsequence of uniformly tight measures)**.** *Let ν_n be positive Borel measures on $\mathbb{R}^2$ with the same finite mass m, and suppose*

$$\lim_{A\to\infty}\sup_n \nu_n(\{x : |x| > A\}) = 0.$$

There are a subsequence and a positive Borel measure ν of mass m such that

$$\int g\,\mathrm{d}\nu_n \longrightarrow \int g\,\mathrm{d}\nu$$

for every bounded continuous function g on $\mathbb{R}^2$.

Proof. The case $m = 0$ is immediate, so divide by m and assume $m = 1$. The coordinatewise map

$$\Phi(x_1, x_2) = \left(\frac{1}{2} + \frac{\arctan x_1}{\pi}, \frac{1}{2} + \frac{\arctan x_2}{\pi}\right)$$

is a homeomorphism onto $(0,1)^2$. Push the measures forward to the compact square $K = [0,1]^2$.

Here is an explicit subsequence construction on K. Encode each coordinate by its binary expansion, taking the expansion ending in zeros at dyadic points and taking all ones at the endpoint one. This is a measurable map from K into $\Omega = (\{0,1\}^2)^{\mathbb{N}}$, and the continuous binary-sum map $\pi : \Omega \to K$ recovers the original point. For each finite word w on the four-symbol alphabet, let $p_n(w)$ be the mass of the cylinder specified by w in the encoded measure. There are countably many such words. A diagonal subsequence makes all $p_n(w)$ converge to numbers $p(w)$. They satisfy

$$p(\varnothing) = 1, \qquad p(w) = \sum_{a\in\{0,1\}^2} p(wa), \qquad p(w) \ge 0.$$

These numbers define a probability on the algebra of finite unions of cylinders, by refining to a common word length.

This algebra probability is a premeasure. Indeed, cylinders are both open and closed for the metric in which agreement in the first k coordinates gives distance at most 2^{-k}. The space Ω is

compact: successive choice of constant coordinates gives a convergent subsequence of any sequence, or equivalently the finitely many length-k cylinders give finite nets and the space is complete. If an algebra set is a disjoint countable union of algebra sets, compactness gives a finite subcover, so only finitely many of the disjoint pieces are nonempty. Finite additivity is therefore already the countable additivity needed on the algebra. The ordinary measure-extension theorem gives a probability P on the Borel sets of Ω. Put $\widetilde{\nu} = \pi_* P$.

For a continuous function g on K, approximate $g \circ \pi$ by a function constant on cylinders of length k, using any point in the corresponding closed dyadic square. The uniform error is at most the modulus of continuity of g at $\sqrt{2}\,2^{-k}$. The integrals of this cylinder function converge because they are finite sums involving $p_n(w)$. First take the subsequence limit and then let $k \to \infty$. This proves convergence of the pushed-forward measures against every continuous function on K.

The measure $\widetilde{\nu}$ gives no mass to ∂K. Given $\delta > 0$, tightness supplies A for which all original measures have mass at most δ outside $[-A, A]^2$. There is a continuous function on K, between zero and one, equal to zero on $\Phi([-A, A]^2)$ and equal to one on ∂K; for example use the quotient of the distance to the first set by the sum of distances to the two disjoint closed sets. Its integrals against the pushed-forward measures are at most δ, and the same is true for $\widetilde{\nu}$. Hence $\widetilde{\nu}(\partial K) \le \delta$. Let $\delta \downarrow 0$.

Pull $\widetilde{\nu}$ back to $(0, 1)^2$ by Φ^{-1} to obtain a probability ν on $\mathbb{R}^2$. Compactly supported continuous tests on $\mathbb{R}^2$ extend by zero to continuous tests on K after this change of coordinates, so their integrals converge. For a bounded continuous test, insert a continuous compact cutoff equal to one on $[-A, A]^2$. The discarded integrals are uniformly small for the original measures by tightness, and small for ν because its finite mass is exhausted by these squares. This proves convergence for every bounded continuous test and finishes the proof. □

Theorem 2.17 (Spectral representation in the required form)**.** *Let $c : \mathbb{R}^2 \to \mathbb{C}$ be a Lipschitz positive-definite function. There is a unique finite positive Borel measure ν such that*

$$c(s) = \int \exp(2\pi i s \cdot \xi)\, \mathrm{d}\nu(\xi) \qquad (s \in \mathbb{R}^2), \qquad \nu(\mathbb{R}^2) = c(0).$$

In particular, $c(0) \le 1$ implies that the representing measure has mass at most one.

Proof. Put $m = c(0)$. The elementary part of Lemma 2.15 gives $m \ge 0$ and $|c| \le m$; if $m = 0$ there is nothing to prove. For $0 < \varepsilon \le 1$ put

$$c_\varepsilon(s) = c(s) \exp(-\pi\varepsilon|s|^2).$$

This function is integrable and positive definite. To see the latter directly, the Gaussian formula represents $\exp(-\pi\varepsilon|s_a - s_b|^2)$ as the integral of $\exp(2\pi i(s_a - s_b) \cdot \xi)$ against the positive probability density $\gamma_\varepsilon(\xi)$. Inside that integral the positive-definite inequality for c_ε is the one for c with coefficients $z_a \exp(2\pi i s_a \cdot \xi)$.

By Lemma 2.15,

$$\mathrm{d}\nu_\varepsilon(\xi) = \widehat{c_\varepsilon}(\xi)\, \mathrm{d}\xi$$

is a positive measure of mass m, with characteristic function c_ε. These measures are uniformly tight. Indeed, for $t > 0$ the Gaussian identity gives

$$\begin{aligned} \int (1 - \exp(-\pi t|\xi|^2))\, \mathrm{d}\nu_\varepsilon(\xi) &= \int \gamma_t(s)(m - \operatorname{Re} c_\varepsilon(s))\, \mathrm{d}s \\ &\le \int \gamma_t(s)(L|s| + m\pi|s|^2)\, \mathrm{d}s \\ &\le C(L\sqrt{t} + mt), \end{aligned}$$

where L is a Lipschitz constant for c. In the second line we used $1 - \exp(-\pi\varepsilon|s|^2) \le \pi|s|^2$ and $|c(s) - c(0)| \le L|s|$. The last line follows by scaling in the first and second Gaussian moments. Set

$t = A^{-2}$. On $|\xi| \geq A$, the integrand on the left is at least $1 - \exp(-\pi)$. Consequently

$$\sup_{0<\varepsilon\leq 1} \nu_\varepsilon(\{|\xi| \geq A\}) \leq C(L/A + m/A^2) \longrightarrow 0.$$

Apply Lemma 2.16 to $\nu_{1/n}$. The limiting measure ν has mass m. For each s, the character $\xi \mapsto \exp(2\pi i s \cdot \xi)$ is bounded and continuous, so

$$\int \exp(2\pi i s \cdot \xi)\, \mathrm{d}\nu(\xi) = \lim_n c(s) \exp(-\pi |s|^2/n) = c(s).$$

Uniqueness follows from Lemma 2.6. □

Lemma 2.18 (Measurable dependence of spectral measures)**.** *Suppose $z \mapsto c_z(s)$ is measurable for each $s \in \mathbb{R}^2$, and every c_z is Lipschitz and positive definite with $c_z(0) \leq 1$. Let ν_z be the measures supplied by Theorem 2.17. Then $z \mapsto \nu_z(B)$ is measurable for every Borel set $B \subset \mathbb{R}^2$.*

Proof. The functions $(z, s) \mapsto c_z(s)$ are jointly measurable: replace s by a sequence of rational-grid approximations and use continuity in s. For $\varphi \in \mathcal{S}$, Fourier inversion and Fubini give

$$\int \varphi(\xi)\, \mathrm{d}\nu_z(\xi) = \int \widehat{\varphi}(s) c_z(s)\, \mathrm{d}s.$$

The right side is measurable in z, since $|c_z(s)| \leq c_z(0) \leq 1$ and $\widehat{\varphi} \in L^1$.

For an open set O, choose smooth compactly supported functions $0 \leq \varphi_m \leq 1$ in O increasing pointwise to $\mathbf{1}_O$. One explicit construction takes cutoffs η_m equal to one on a compact exhaustion of O, with support inside O, and sets $\varphi_m = 1 - \prod_{j=1}^m (1 - \eta_j)$. Thus $\nu_z(O) = \lim_m \int \varphi_m \, \mathrm{d}\nu_z$ is measurable. The class of sets with measurable ν_z-mass is closed under complements and countable disjoint unions, since the total mass is the measurable function $c_z(0)$. Open sets are closed under finite intersections and generate the Borel sets. The elementary monotone-class argument therefore extends measurability to all Borel sets. □

Definition 2.19 (Measurable measure kernels)**.** Let $(X, \mathcal{A})$ be a measurable space. A family $z \mapsto \nu_z$ of finite positive Borel measures on $\mathbb{R}^q$ is a *measurable kernel* if $z \mapsto \nu_z(B)$ is $\mathcal{A}$-measurable for every Borel set $B \subset \mathbb{R}^q$.

Lemma 2.20 (Integration against measurable kernels)**.** *Let $(X, \mathcal{A}, \lambda)$ be a sigma-finite measure space and let $z \mapsto \nu_z$ be a measurable kernel on $\mathbb{R}^q$. For every nonnegative $\mathcal{A} \otimes \mathcal{B}(\mathbb{R}^q)$-measurable function F, the function $z \mapsto \int F(z, \xi)\, \mathrm{d}\nu_z(\xi)$ is measurable. There is a sigma-finite measure $\mathfrak{m}$ on the product space satisfying*

$$\int F\, \mathrm{d}\mathfrak{m} = \int_X \int_{\mathbb{R}^q} F(z, \xi)\, \mathrm{d}\nu_z(\xi)\, \mathrm{d}\lambda(z).$$

If $z \mapsto \eta_z$ is another measurable finite positive kernel, then $z \mapsto \nu_z \otimes \eta_z$ is a measurable kernel. These identities also hold for integrable complex functions.

Proof. For the indicator of a rectangle $A \times B$, the inner integral is $\mathbf{1}_A(z)\nu_z(B)$ and is measurable. The class of product-measurable sets for which the section integral is measurable is closed under complements, because $\nu_z(\mathbb{R}^q)$ is finite and measurable, and under countable disjoint unions by monotone convergence. Rectangles form a system closed under finite intersections and generate the product sigma-algebra. The monotone-class theorem therefore proves the claim for indicators of all product-measurable sets. Increasing simple approximations prove it for all nonnegative F.

For such a set A, define

$$\mathfrak{m}(A) = \int_X \nu_z(A_z)\, \mathrm{d}\lambda(z), \qquad A_z = \{\xi : (z, \xi) \in A\}.$$

Sections are Borel measurable, and the preceding argument makes this definition meaningful. Countable additivity follows from countable additivity of each ν_z and monotone convergence under the outer integral. If X_n increase to X and $\lambda(X_n) < \infty$, the sets

$$\big(X_n \cap \{z : \nu_z(\mathbb{R}^q) \le n\}\big) \times \mathbb{R}^q$$

exhaust the product and have $\mathfrak{m}$-measure at most $n\lambda(X_n)$. Thus $\mathfrak{m}$ is sigma-finite. The integral identity follows first for indicators from its definition, then for simple functions, and then by monotone convergence.

For product kernels, the mass of a rectangle $B \times C$ is $\nu_z(B)\eta_z(C)$ and is measurable. The same monotone-class argument, using the finite measurable total mass, extends this to every Borel subset of the product Euclidean space. Finally split an integrable complex function into positive and negative real and imaginary parts. Ordinary Tonelli applied to the constructed measure and an additional independent measure permits the angular integrations used below to be interchanged as well. □

2.4. **Local matrix groups and analytic zero sets.**

Lemma 2.21 (A continuous local matrix group is exponential)**.** *Let U be a neighborhood of zero in $\mathbb{R}^2$. Suppose $C : U \to \mathrm{GL}_d(\mathbb{R})$ is continuous, $C(0) = I$, and $C(h + k) = C(h)C(k)$ whenever $h, k, h + k \in U$. There are commuting real matrices T_1, T_2 such that, on a possibly smaller neighborhood of zero,*

$$C(h) = \exp(h_1T_1 + h_2T_2).$$

The expression on the right is real analytic for all $h \in \mathbb{R}^2$.

Proof. For $i = 1, 2$ write $C_i(t) = C(te_i)$ near zero. Choose $a > 0$ sufficiently small that the matrix

$$A_i = \int_0^a C_i(t)\,\mathrm{d}t$$

is invertible: $a^{-1}A_i \to I$ as $a \downarrow 0$, and invertibility is open because the determinant is continuous. For h in a small interval around zero, the local group law gives

$$C_i(h)A_i = \int_h^{h+a} C_i(t)\,\mathrm{d}t.$$

The right side is differentiable even though differentiability of C_i was not assumed. Multiplying by A_i^{-1} first shows that C_i is differentiable, and differentiating gives

$$C_i'(h) = C_i(h)T_i, \qquad T_i = (C_i(a) - I)A_i^{-1}.$$

The derivative of $C_i(h)\exp(-hT_i)$ is zero. Its value at zero is I, so $C_i(h) = \exp(hT_i)$ near zero. The matrix exponential and its derivative here are obtained directly from its absolutely convergent power series.

Commutativity of addition and the local group law yield

$$\exp(sT_1)\exp(tT_2) = \exp(tT_2)\exp(sT_1)$$

for small s, t. Differentiate at $s = t = 0$ to get $T_1T_2 = T_2T_1$. Hence

$$C(h) = C(h_1e_1)C(h_2e_2) = \exp(h_1T_1)\exp(h_2T_2) = \exp(h_1T_1 + h_2T_2).$$

The last equality follows by multiplying the exponential series and using the binomial formula for commuting matrices. Those series converge locally uniformly with all derivatives, which also proves real analyticity on the whole plane. □

Lemma 2.22 (Zero sets of real analytic functions)**.** *Let $F : U \to \mathbb{R}$ be real analytic on a connected open set $U \subset \mathbb{R}^d$, where $d = 1, 2,$ or 3. If F vanishes on a set of positive Lebesgue measure, it vanishes identically. In particular, a real analytic function on $\mathbb{R}^2 \times (\mathbb{R}/(2\pi\mathbb{Z}))$ which vanishes on a set of positive product measure is identically zero.*

Proof. Let Z_∞ be the set of points at which all partial derivatives of F, including F itself, vanish. This set is closed in U by continuity of the derivatives. It is also open: the convergent Taylor series at one of its points is zero, so F is identically zero in a neighborhood, where all of its derivatives vanish as well. Connectedness shows that Z_∞ is either empty or all of U. In the latter case the assertion is proved.

Suppose therefore that F is not identically zero. At every point x some Taylor coefficient is nonzero. If the first nonzero homogeneous term has degree k, it takes a nonzero value in some direction v, so $\partial_v^k F(x) \neq 0$. For $k = 0$ this simply means $F(x) \neq 0$. By continuity there is a ball about x on which this derivative never vanishes. On every line parallel to v intersecting the ball, the restriction of F has at most k zeros. Otherwise repeated Rolle's theorem would give a zero of its kth derivative on that interval. Fubini, after an orthogonal change of coordinates, therefore shows that the zero set of F in the ball has d-dimensional measure zero. A countable subcover of these balls covers U, so the entire zero set has measure zero, contradicting the hypothesis.

For the final assertion, lift the periodic angular variable to $\theta \in \mathbb{R}$. This gives a real analytic function on the connected space $\mathbb{R}^3$. A positive-measure zero set in the quotient has a positive-measure lift in one angular period. Apply the result just proved to the lifted function. □

3. Finite words, binary density, and tilings

The elementary combinatorial material in this section supplies the binary base case and the tiling step of the density argument. The organization of the tiling proof follows the uniform-measure argument of Dodos, Kanellopoulos, and Tyros [3]; every ingredient needed here is proved below.

Definition 3.1 (Words, subspaces, and uniform density)**.** For a positive integer k, let $[k] = \{1, \dots, k\}$. For $k \geq 2$ and $n \geq 0$, let $[k]^n$ be the set of words of length n over $[k]$; $[k]^0$ consists of the empty word. Write $\mu_{k,n}(A) = |A|/k^n$ for uniform density. The concatenation of words x and y is denoted by $x \frown y$.

An m-dimensional combinatorial subspace of $[k]^n$ is specified by pairwise disjoint, nonempty coordinate sets $I_1, \dots, I_m$ and by fixed letters on their complement. Its coordinate map $\phi : [k]^m \longrightarrow [k]^n$ puts the letter a_j on every coordinate in I_j and the specified fixed letters elsewhere. The subspace is $V = \phi([k]^m)$. We order the variable sets by their least coordinates when a definite choice of coordinate map is required. A one-dimensional subspace is called a *combinatorial line*. For $A \subseteq [k]^n$, put $\mu_V(A) = |A \cap V|/|V|$. If $V \subseteq [3]^n$, its *binary face* is $V^{(2)} = \phi([2]^m)$; fixed letters of V may still equal 3. For $x \in [k]^\ell$, the suffix fiber of $A \subseteq [k]^n$ at x is

$$A_x = \{y \in [k]^{n-\ell} : x \frown y \in A\}.$$

Lemma 3.2 (Subspace bookkeeping)**.** *An m-dimensional subspace has k^m points. The image of an r-dimensional subspace under the coordinate map of an m-dimensional subspace is an r-dimensional subspace. Concatenating subspaces of dimensions r and s gives a subspace of dimension $r+s$. There are at most $(k+m)^n$ m-dimensional subspaces of $[k]^n$ and at most $(k+1)^n$ lines.*

Proof. Each variable set is nonempty, so its common letter can be read from any one of its coordinates. Thus the coordinate map is injective. Under composition, each new variable set is a union of distinct old variable sets; these unions remain nonempty and disjoint. Fixed parameters simply turn the corresponding old variable sets into fixed coordinates. Concatenation places the two collections of variable sets in disjoint coordinate intervals. Finally, encode a subspace by a word over $[k] \cup \{v_1, \dots, v_m\}$, with v_j marking its jth variable set. Each subspace has such an encoding among the $(k+m)^n$ possible words. Taking $m = 1$ gives the line bound. □

Lemma 3.3 (Binary density theorem)**.** *For $0 < \alpha \leq 1$, set*

$$H_2(\alpha) = 2^{\lceil \alpha^{-2} \rceil}.$$

If $n \geq H_2(\alpha)$ and $A \subseteq [2]^n$ has $\mu_{2,n}(A) \geq \alpha$, then A contains a combinatorial line.

Proof. Identify a word x with the set $S(x)$ of coordinates carrying the letter 2. Two distinct words form a binary combinatorial line exactly when one of their associated sets is contained in the other. Indeed, the variable set of the line is the difference of the two sets. A line-free family therefore corresponds to an antichain $\mathcal{A}$ of subsets of $[n]$.

Choose a uniformly random permutation of $[n]$ and take its chain of initial segments, including the empty set and $[n]$. The probability that this chain contains a specified j-element set is $\binom{n}{j}^{-1}$. A chain meets an antichain at most once, so taking expectations gives

$$\sum_{S\in\mathcal{A}} \binom{n}{|S|}^{-1} \le 1, \qquad |\mathcal{A}| \le \binom{n}{\lfloor n/2\rfloor}.$$

The binomial coefficients are maximal at the middle because $\binom{n}{j+1}/\binom{n}{j} = (n-j)/(j+1)$. For $n = 2r$, their normalized maximum satisfies

$$\frac{\binom{2r}{r}}{2^{2r}} = \prod_{j=1}^{r}\left(1-\frac{1}{2j}\right) \le \exp\left(-\frac{1}{2}\sum_{j=1}^{r}\frac{1}{j}\right) \le (r+1)^{-1/2}.$$

Here $1-t \le e^{-t}$ and $\sum_{j=1}^{r} j^{-1} \ge \int_1^{r+1} t^{-1}\,\mathrm{d}t$; the displayed bound is also valid when $r = 0$. For $n = 2r+1$, the normalized maximum is $(2r+1)/(2r+2)$ times that for $2r$, and is no larger. Consequently a line-free subset of $[2]^n$ has density at most $(\lfloor n/2\rfloor+1)^{-1/2}$. When $n \ge 2\lceil \alpha^{-2}\rceil$, this last quantity is strictly less than α, a contradiction. □

Lemma 3.4 (Multidimensional binary density theorem)**.** *For every integer $m \ge 1$ and $0 < \alpha \le 1$, there is an integer $B(m,\alpha)$ such that every $A \subseteq [2]^n$ with $n \ge B(m,\alpha)$ and $\mu_{2,n}(A) \ge \alpha$ contains an m-dimensional subspace. One possible recursive choice is*

$$B(1,\alpha) = H_2(\alpha),$$
$$M = H_2(\alpha/2),$$
$$B(m+1,\alpha) = M + B\left(m, \frac{\alpha}{2\,3^M}\right).$$

Proof. The case $m = 1$ is Lemma 3.3. Suppose the assertion is known for m, and split a word of length n into a prefix x of length $n-M$ and a suffix of length M. The average suffix-fiber density is at least α. The prefixes whose fibers have density at least $\alpha/2$ have density at least $\alpha/2$: if their density is p, the average is at most $p + \alpha/2$. Every such fiber contains a line by the choice of M. Choose one line from each of these fibers. There are at most 3^M possible lines by Lemma 3.2, so some fixed line $L \subseteq [2]^M$ lies in every fiber above a set C of prefixes of density at least $\alpha/(2\,3^M)$. The induction hypothesis gives an m-dimensional subspace $U \subseteq C$. Then $U \frown L \subseteq A$ is an $(m+1)$-dimensional subspace. □

Lemma 3.5 (Uniform lower bounds for prefix fibers)**.** *For $k \ge 2$, $m \ge 1$, and $0 < \varepsilon < 1$, define*

$$P(k,m,\varepsilon) = m\left(\left\lceil \frac{k^m-1}{\varepsilon}\right\rceil + 2\right).$$

If $n \ge P(k,m,\varepsilon)$ and $A \subseteq [k]^n$, there are $\ell < n$ and an m-dimensional subspace $U \subseteq [k]^\ell$ such that

$$\mu_{k,n-\ell}(A_u) \ge \mu_{k,n}(A) - \varepsilon \qquad (u \in U).$$

Proof. Put $\alpha = \mu_{k,n}(A)$, $K = k^m$, and $\rho = \varepsilon/(K-1)$; let $J = \lceil 1/\rho\rceil$. We construct a fixed prefix a block of m coordinates at a time. Initially the prefix is empty and its suffix fiber has density α. At stage j, suppose the chosen prefix has length jm and its remaining fiber has density $a_j \ge \alpha + j\rho$. Split that remaining fiber according to its next m coordinates. There are K child fibers whose densities average to a_j. If every child has density at least $\alpha - \varepsilon$, the desired U is obtained by appending all $[k]^m$ to the current prefix, and we stop.

Otherwise one child has density less than $\alpha - \varepsilon \le a_j - \varepsilon$. The sum of the other $K-1$ densities is therefore greater than $(K-1)a_j + \varepsilon$. At least one of those children has density greater than $a_j + \rho$. Select its m-letter word as the next fixed block. This establishes the induction hypothesis for stage $j+1$.

The process cannot fail to stop at all stages $0, \dots, J$, since $J+1$ failures would produce a fiber of density greater than $\alpha + (J+1)\rho > 1$. The assumed dimension permits all these stages: at stage $j \le J$ the length $\ell = (j+1)m$ is at most $(J+1)m < n$. Thus the terminal prefix subspace and all of its suffix fibers are well defined and have the stated property. □

Lemma 3.6 (A binary face in a dense ternary cube)**.** *For $d \ge 1$ and $0 < \alpha \le 1$, put*

$$M = B(d, \alpha/2), \qquad F_2(d, \alpha) = P(3, M, \alpha/2).$$

If $n \ge F_2(d, \alpha)$ and $A \subseteq [3]^n$ has $\mu_{3,n}(A) \ge \alpha$, there is a d-dimensional ternary subspace V such that $V^{(2)} \subseteq A$.

Proof. By Lemma 3.5, there are $\ell < n$ and an M-dimensional $U \subseteq [3]^\ell$ such that every suffix fiber above U has density at least $\alpha/2$. The same is true above $U^{(2)}$. Averaging first over $u \in U^{(2)}$ and then over suffixes shows that some suffix y satisfies

$$\frac{|\{u \in U^{(2)} : u \frown y \in A\}|}{2^M} \ge \alpha/2.$$

Via the coordinate map of U, this is a subset of $[2]^M$ of density at least $\alpha/2$. Lemma 3.4 gives a d-dimensional binary subspace in this subset. Its encoding uses fixed letters $1, 2$ and d variables. Allow those variables also to take the letter 3, compose with the coordinate map of U, and append y. Lemma 3.2 shows that the result is a d-dimensional ternary subspace, and its binary face is contained in A by construction. □

Definition 3.7 (Insensitive sets)**.** For $i \in \{1, 2\}$, two words in $[3]^n$ are $(i, 3)$-*equivalent* if, at each coordinate, either their letters are equal or both letters belong to $\{i, 3\}$. A set $D \subseteq [3]^n$ is $(i, 3)$-*insensitive* if it is a union of these equivalence classes. A subset of a subspace is called $(i, 3)$-insensitive in that subspace when its inverse image under the coordinate map is $(i, 3)$-insensitive.

Lemma 3.8 (Elementary properties of insensitivity)**.** *Unions, intersections, and complements of $(i, 3)$-insensitive sets are $(i, 3)$-insensitive. Fixing any collection of coordinates preserves insensitivity on the remaining coordinates. The inverse image of an $(i, 3)$-insensitive set under any subspace coordinate map is $(i, 3)$-insensitive. Finally, if D is $(i, 3)$-insensitive and $V^{(2)} \subseteq D$, then $V \subseteq D$.*

Proof. The first assertions follow because membership is constant on each equivalence class. Fixing coordinates restricts that equivalence relation to the unfixed coordinates. For a subspace coordinate map, changing a parameter between i and 3 changes all coordinates of its variable set between those same two letters, so equivalent parameter words have equivalent images. For the final assertion, take a point of V and replace every parameter equal to 3 by i. The resulting point lies in $V^{(2)}$ and is $(i, 3)$-equivalent to the original point. Its membership in D therefore implies membership of the original point. □

Lemma 3.9 (Tiling one insensitive set)**.** *For every $d \ge 1$ and $0 < \varepsilon \le 1$ there is an integer $T_1(d, \varepsilon)$ with the following property. If $n \ge T_1(d, \varepsilon)$, $i \in \{1, 2\}$, and $D \subseteq [3]^n$ is $(i, 3)$-insensitive, there is a finite family $\mathcal{V}$ of pairwise disjoint d-dimensional subspaces, all contained in D, such that*

$$\mu_{3,n}\left(D \setminus \bigcup_{V \in \mathcal{V}} V\right) \le \varepsilon.$$

A valid choice is obtained from

$$\beta = \varepsilon/2, \quad M = F_2(d, \beta), \quad Q = (3+d)^M, \quad \vartheta = \frac{\beta}{Q}\, 3^{d-M}$$

by setting

$$L = \lceil \vartheta^{-1} \rceil + 1, \qquad T_1(d, \varepsilon) = LM.$$

Proof. We remove subspaces greedily, using successive blocks of M coordinates from right to left. The invariant is the following: after t blocks have been used, the unremoved set $R_t \subseteq D$ has the property that, for every suffix $y \in [3]^{tM}$, its prefix section

$$(R_t)^y = \{x \in [3]^{n-tM} : x \frown y \in R_t\}$$

is $(i, 3)$-insensitive on its $n - tM$ coordinates. This holds for $t = 0$, with $R_0 = D$.

If $\mu_{3,n}(R_t) \le \varepsilon$, stop. Otherwise assume $t < L$ and write a word as $x \frown z \frown y$, with respective lengths $n - (t+1)M$, M, and tM. For a pair (x, y), put

$$E_{x,y} = \{z \in [3]^M : x \frown z \frown y \in R_t\}.$$

The invariant and Lemma 3.8 imply that $E_{x,y}$ is $(i, 3)$-insensitive. Its average density, as (x, y) ranges uniformly over the indicated product, is $\mu_{3,n}(R_t) > 2\beta$. Hence the pairs with $\mu_{3,M}(E_{x,y}) \ge \beta$ have product density at least β: if that density is p, the average is at most $p + \beta$.

For each such pair, Lemma 3.6 supplies a d-dimensional subspace whose binary face lies in $E_{x,y}$. Lemma 3.8 puts the entire subspace in $E_{x,y}$. There are at most Q possible subspaces by Lemma 3.2. Consequently some fixed d-dimensional $W \subseteq [3]^M$ has

$$S = \{(x, y) : x \frown W \frown y \subseteq R_t\}$$

of product density at least β/Q. Remove every set $x \frown W \frown y$ for $(x, y) \in S$. These sets are pairwise disjoint d-dimensional subspaces: different pairs have different fixed prefix or suffix, and W varies only in the middle block. They are all in R_t, so they are disjoint from all subspaces removed at earlier stages. The density removed at this stage is

$$\frac{|S| 3^d}{3^n} = \frac{|S|}{3^{n-M}} 3^{d-M} \ge \vartheta.$$

It remains to check the invariant; the residual need not be insensitive on *all* its coordinates. For a fixed old suffix y, define

$$S_y = \{x : (x, y) \in S\} = \bigcap_{z \in W} \{x : x \frown z \frown y \in R_t\}.$$

Each set in this intersection is an insensitive prefix section by the induction hypothesis, so S_y is insensitive. For any fixed new suffix $z \frown y$, the section of R_{t+1} is

$$\{x : x \frown z \frown y \in R_{t+1}\} = \begin{cases} \{x : x \frown z \frown y \in R_t\} \setminus S_y, & z \in W, \\ \{x : x \frown z \frown y \in R_t\}, & z \notin W. \end{cases}$$

Both alternatives are $(i, 3)$-insensitive. This proves the invariant for the next stage, including when the remaining prefix has length zero.

All L stages would be possible dimensionally, since $n \ge LM$. If the algorithm performed L removal stages without stopping, it would remove total density at least $L\vartheta > 1$, which is impossible. It therefore stops with residual density at most ε. The collection of all removed subspaces is the required finite family. If the initial density is already at most ε, the empty family suffices. □

Lemma 3.10 (Tiling an intersection of two insensitive sets)**.** *For $d \ge 1$ and $0 < \varepsilon \le 1$, put*

$$M = T_1(d, \varepsilon/2), \qquad T_2(d, \varepsilon) = T_1(M, \varepsilon/2).$$

If $n \ge T_2(d, \varepsilon)$, $D_1 \subseteq [3]^n$ is $(1, 3)$-insensitive, and $D_2 \subseteq [3]^n$ is $(2, 3)$-insensitive, then $D = D_1 \cap D_2$ contains a finite family $\mathcal{V}$ of pairwise disjoint d-dimensional subspaces such that

$$\mu_{3,n}\left(D \setminus \bigcup_{V \in \mathcal{V}} V\right) \le \varepsilon.$$

Proof. Apply Lemma 3.9 to D_1, with target dimension M and error $\varepsilon/2$. This gives pairwise disjoint M-dimensional subspaces $W \subseteq D_1$ whose union misses at most $\varepsilon/2$ of the ambient density of D_1. Inside each W, the pullback of $D_2 \cap W$ is $(2,3)$-insensitive by Lemma 3.8. Its ambient dimension is $M = T_1(d, \varepsilon/2)$, so a second application of Lemma 3.9 tiles it by d-dimensional subspaces, leaving relative density at most $\varepsilon/2$ in W.

All resulting subspaces are contained in $D_1 \cap D_2$ and are pairwise disjoint, since the outer subspaces W are disjoint. The part of D outside their union consists of a subset of the uncovered portion of D_1, together with the residuals inside the outer subspaces. Its ambient density is therefore at most

$$\frac{\varepsilon}{2} + \sum_W \mu_{3,n}(W)\,\frac{\varepsilon}{2} \le \varepsilon.$$

This proves the assertion. □

3.1. **Equal-slices averaging and the three-letter theorem.** The remaining argument specializes the density-increment method of [2] to three letters. Its two changes of probability measure are proved below by direct counting. Thus no Ramsey theorem for parameter sets, and no density Hales–Jewett theorem, is used as an input.

Definition 3.11 (Equal-slices probability and total variation)**.** Retain the uniform probability $\mu_{k,n}$ from Definition 3.1. For $k \in \{2,3\}$ and $n \ge 0$, the *equal-slices probability* $\nu_{k,n}$ is obtained by first choosing uniformly a vector $(a_1, \dots, a_k)$ of nonnegative integers with $a_1 + \cdots + a_k = n$, and then choosing uniformly a word having exactly a_j occurrences of the letter j for every j. For $n = 0$ both measures give mass one to the empty word.

For probabilities P, Q on a finite set X, put

$$d_{\mathrm{TV}}(P,Q) = \frac{1}{2}\sum_{x \in X} |P(\{x\}) - Q(\{x\})|.$$

For integers $a \ge 0$ and $r \ge 0$, the falling factorial is $(a)_r = a(a-1)\cdots(a-r+1)$, with $(a)_0 = 1$ and $(a)_r = 0$ if $r > a$.

Lemma 3.12 (Elementary counting and total variation)**.** *If $x \in [k]^n$ has a_j occurrences of j, then*

$$\nu_{k,n}(\{x\}) = \binom{n+k-1}{k-1}^{-1} \binom{n}{a_1, \dots, a_k}^{-1} = \frac{(k-1)!\,a_1! \cdots a_k!}{(n+k-1)!}. \tag{3.1}$$

In particular,

$$\nu_{3,n}([2]^n) = \frac{2}{n+2}. \tag{3.2}$$

For any $0 \le h \le 1$ on a finite set,

$$|\mathbb{E}_P h - \mathbb{E}_Q h| \le d_{\mathrm{TV}}(P,Q). \tag{3.3}$$

Moreover, if $P = \sum_j a_j P_j$, where $a_j \ge 0$ and $\sum_j a_j = 1$, then

$$d_{\mathrm{TV}}(P,Q) \le \sum_j a_j d_{\mathrm{TV}}(P_j, Q).$$

Proof. A vector of k nonnegative integers summing to n is specified by placing $k-1$ separators among a row of n indistinguishable marks. There are $\binom{n+k-1}{k-1}$ possibilities. There are $n!/(a_1! \cdots a_k!)$ words with a prescribed count vector, proving (3.1). Exactly $n+1$ of the $\binom{n+2}{2}$ count vectors for three letters have $a_3 = 0$; this gives (3.2).

The signed masses $P(\{x\}) - Q(\{x\})$ sum to zero. Their positive sum and the absolute value of their negative sum are therefore both $d_{\mathrm{TV}}(P,Q)$. Multiplication by numbers in $[0,1]$ makes the resulting sum lie between the negative and positive sums. This proves (3.3). The last assertion follows by applying the triangle inequality inside the defining finite sum. □

Definition 3.13 (Random coordinate restriction)**.** Let $n \geq r$. Choose an injection $\iota : [r] \longrightarrow [n]$ uniformly among the $(n)_r$ injections. For a word y on the complement of its image, define

$$\phi_{\iota,y} : [k]^r \longrightarrow [k]^n$$

by putting the jth input letter in coordinate $\iota(j)$ and putting the prescribed letters y in all remaining coordinates. Its image is a coordinate subspace, hence a combinatorial subspace in the sense of Definition 3.1. For $A \subseteq [k]^n$, write $A_{\iota,y} = \phi_{\iota,y}^{-1}(A)$.

Lemma 3.14 (Inserting a fixed number of letters into uniform background)**.** *Fix $k \in \{2,3\}$, $r \geq 1$, and $\zeta > 0$. There is $N = N(k,r,\zeta) \geq r$ such that the following holds for every $n \geq N$. Choose ι as in Definition 3.13, give the remaining coordinates independent uniform letters, and independently choose $w \in [k]^r$ with any probability distribution λ. The law of $\phi_{\iota,y}(w)$ is within ζ in total variation of $\mu_{k,n}$. The same N works for every λ.*

Proof. First take λ to be concentrated on a single word w, and let c_j be the number of its letters equal to j. For an output word z, let a_j be the number of its letters equal to j. There are $\prod_j (a_j)_{c_j}$ injections whose specified letters agree with z. Once the injection has been chosen, the probability that the remaining uniform letters agree with z is $k^{-(n-r)}$. The density of the output law with respect to $\mu_{k,n}$ is consequently

$$L_{n,w}(z) = k^r \frac{\prod_{j=1}^k (a_j)_{c_j}}{(n)_r}. \tag{3.4}$$

It lies in $[0, k^r]$, because the injections counted in the numerator are some of the $(n)_r$ possible injections.

For a uniform random z, the random variable a_j is the sum of n independent indicators, each having mean $1/k$. Expanding the square and using independence gives

$$\mathbb{E}a_j = \frac{n}{k}, \qquad \mathbb{E}\left(a_j - \frac{n}{k}\right)^2 = n\frac{1}{k}\left(1 - \frac{1}{k}\right).$$

For every $\tau > 0$, the elementary inequality $\mathbf{1}_{\{|X|>\tau\}} \leq |X|^2/\tau^2$ therefore yields

$$\mathbb{P}\left(\left|\frac{a_j}{n} - \frac{1}{k}\right| > \tau\right) \leq \frac{k-1}{k^2 n \tau^2}.$$

Thus all a_j/n approach $1/k$ in probability. Since r and the c_j are fixed, divide the numerator and denominator of (3.4) by n^r to see that $L_{n,w}$ approaches 1 in probability. This also follows directly by writing each factor as $a_j/n - s/n$ and each denominator factor as $1 - s/n$. For any $u > 0$,

$$\mathbb{E}_{\mu_{k,n}}|L_{n,w} - 1| \leq u + (k^r + 1)\,\mathbb{P}_{\mu_{k,n}}(|L_{n,w} - 1| > u).$$

First let n increase and then let u decrease to zero. It follows that the total variation distance, which is one half of this expectation, tends to zero.

There are only k^r choices of w. Choose a single N that works for every one of them. Any distribution λ gives a mixture of these output laws, so the mixture inequality of Lemma 3.12 proves the assertion. □

Lemma 3.15 (Inserting letters into equal-slices background)**.** *Let $m \geq r \geq 1$. Choose $\iota : [r] \to [m]$ uniformly, choose y on the remaining $m - r$ coordinates with probability $\nu_{3,m-r}$, and independently choose $w \in [3]^r$ with any distribution. If P is the law of $\phi_{\iota,y}(w)$, then*

$$d_{\mathrm{TV}}(P, \nu_{3,m}) \leq 1 - \frac{(m-r+1)(m-r+2)}{(m+1)(m+2)} \leq \frac{2r}{m+1} \leq \frac{3r}{m}. \tag{3.5}$$

In particular, if w is uniform, then for every $H \subseteq [3]^m$,

$$|\mathbb{E}_{\iota,y}\mu_{3,r}(H_{\iota,y}) - \nu_{3,m}(H)| \leq \frac{3r}{m}. \tag{3.6}$$

Proof. Fix w, with letter counts c_1, c_2, c_3, and let z have letter counts a_1, a_2, a_3. If any $a_j < c_j$, the output cannot be z. Otherwise its probability is

$$\frac{\prod_{j=1}^{3}(a_j)_{c_j}}{(m)_r}\,\frac{2!\,\prod_{j=1}^{3}(a_j-c_j)!}{(m-r+2)!}.$$

Divide this by the equal-slices atom from (3.1). All the factorials depending on the a_j cancel. The resulting density is

$$L_w(z) = C_{m,r}\,\mathbf{1}_{\{a_j \ge c_j \text{ for all } j\}}, \qquad C_{m,r} = \frac{(m+2)_r}{(m)_r} = \frac{(m+1)(m+2)}{(m-r+1)(m-r+2)}.$$

Since this density has integral one, the indicated event has $\nu_{3,m}$-probability $C_{m,r}^{-1}$. Its total variation distance from $\nu_{3,m}$ is exactly $1 - C_{m,r}^{-1}$: the positive part of $L_w - 1$ and the negative part have that same integral. The bound for arbitrary w-distributions follows by taking mixtures. Finally,

$$1 - C_{m,r}^{-1} = \frac{r(2m-r+3)}{(m+1)(m+2)} \le \frac{2r}{m+1}.$$

Equation (3.6) is (3.3) applied to the indicator of H. □

Definition 3.16 (Collapsing the third letter)**.** For $i \in \{1,2\}$ and $x \in [3]^m$, let $x^{3\to i} \in [2]^m$ be obtained by replacing every 3 by i. For $B \subseteq [2]^m$, put

$$C_i(B) = \{x \in [3]^m : x^{3\to i} \in B\}, \qquad C(B) = C_1(B) \cap C_2(B).$$

Lemma 3.17 (A quantitative supply of binary pairs)**.** *Let $B \subseteq [2]^m$ and $b = \nu_{2,m}(B)$. Each $C_i(B)$ is $(i,3)$-insensitive, and*

$$\nu_{3,m}(C(B)) \ge \frac{b^2}{2}. \tag{3.7}$$

If $A \subseteq [3]^m$ contains no combinatorial line and $B = A \cap [2]^m$, then

$$A \cap C(B) \subseteq [2]^m. \tag{3.8}$$

Proof. Changing any occurrences of i and 3 leaves $x^{3\to i}$ unchanged. This proves the asserted insensitivity.

Choose a uniform permutation π of $[m]$. For $0 \le s \le m$, let $u_s(\pi)$ have letter 1 at $\pi(1), \dots, \pi(s)$ and letter 2 elsewhere. If s is chosen uniformly from $\{0, \dots, m\}$, then $u_s(\pi)$ has law $\nu_{2,m}$: its number of ones is uniform, and conditional on that number their positions are uniform. Set

$$h(\pi) = \frac{1}{m+1}\sum_{s=0}^{m} \mathbf{1}_B(u_s(\pi)).$$

Then $\mathbb{E}_\pi h = b$. For independent uniform ranks s, t,

$$\mathbb{P}(u_s(\pi), u_t(\pi) \in B) = \mathbb{E}_\pi h(\pi)^2 \ge b^2,$$

because $\mathbb{E}(h-b)^2 \ge 0$.

Put $a = \min(s,t)$ and $c = \max(s,t)$. Make a ternary word z by giving the first a positions in the order π letter 1, the next $c-a$ positions letter 3, and the remaining positions letter 2. Its two collapsed words are precisely $u_a(\pi)$ and $u_c(\pi)$. Hence the probability that $z \in C(B)$ is at least b^2.

For $a < c$, the count vector $(a, m-c, c-a)$ has probability $2/(m+1)^2$ under this construction; for $a = c$ it has probability $1/(m+1)^2$. Under $\nu_{3,m}$ every count vector has probability $2/((m+1)(m+2))$. Conditional on a count vector, both constructions are uniform on its words. Therefore the probability of every word under the constructed law is at most $(m+2)/(m+1)$ times its equal-slices probability. Consequently

$$\nu_{3,m}(C(B)) \ge \frac{m+1}{m+2}b^2 \ge \frac{b^2}{2}.$$

Finally, if $z \in A \cap C(B)$ has at least one 3, the words $z^{3\to1}, z^{3\to2}, z$ form a combinatorial line, with variable set equal to the positions of the 3's. All three words belong to A, a contradiction. This proves (3.8). □

Proposition 3.18 (A three-letter density increment)**.** *For every $0 < \delta \le 1$ there is $h(\delta) > 0$ with the following property. For every integer $d \ge 1$ there is an integer $N(d,\delta) \ge d$ such that, whenever $n \ge N(d,\delta)$ and $A \subseteq [3]^n$ has uniform density $\alpha \ge \delta$ and contains no combinatorial line, some d-dimensional combinatorial subspace V satisfies*

$$\frac{|A \cap V|}{|V|} \ge \alpha + h(\delta).$$

One may take $h(\delta) = \delta^4/65536$.

Proof. Fix δ and d. Set

$$\theta = \frac{\delta^2}{64}, \qquad t = \frac{\delta\theta}{8}, \qquad \kappa = \frac{t}{8}, \qquad \varepsilon = \frac{\delta\kappa}{8}, \qquad h = \frac{\varepsilon}{2} = \frac{\delta^4}{65536}. \tag{3.9}$$

In particular, $\kappa \le \delta/4$ and $\varepsilon \le \delta/4$.

Use Lemma 3.10 to choose $r \ge d$ so that every intersection of a $(1,3)$-insensitive set and a $(2,3)$-insensitive set in $[3]^r$ can be covered by disjoint d-dimensional combinatorial subspaces contained in the intersection, with uncovered uniform measure at most $t/8$. Choose $m \ge 3r$ so large that

$$\frac{3r}{m} \le \min\left(\frac{\varepsilon}{2}, \frac{t}{8}\right), \qquad \frac{2}{m+2} \le \frac{\delta\theta}{2}. \tag{3.10}$$

Finally choose $N(d,\delta) \ge m$ so that Lemma 3.14, with $k = 3$, $r = m$ and error ε, holds for all $n \ge N(d,\delta)$. All these choices depend on d and δ, and not on A or its actual density α.

Let $n \ge N(d,\delta)$ and let A satisfy the hypotheses. Choose a random coordinate restriction $\phi : [3]^m \to [3]^n$ with uniform letters in the fixed coordinates, and put

$$a(\phi) = \nu_{3,m}(\phi^{-1}(A)), \qquad b(\phi) = \nu_{2,m}(\phi^{-1}(A) \cap [2]^m).$$

Apply Lemma 3.14 first with inserted-word law $\nu_{3,m}$ and then with inserted-word law $\nu_{2,m}$, viewed as a probability on $[3]^m$ supported on $[2]^m$. It gives

$$\mathbb{E}a \ge \alpha - \varepsilon, \qquad \mathbb{E}b \ge \alpha - \varepsilon. \tag{3.11}$$

Suppose first that some restriction ϕ has $a(\phi) \ge \alpha + \varepsilon$. Inside its copy of $[3]^m$, choose a random r-coordinate restriction with equal-slices background, as in Lemma 3.15. By (3.6) and (3.10), the average uniform density of $\phi^{-1}(A)$ on these r-dimensional subspaces is at least $\alpha + \varepsilon/2$. One of them has at least this density. Partition that subspace by fixing $r - d$ of its free coordinates. The resulting d-dimensional subspaces have equal size, so one has density at least $\alpha + \varepsilon/2 = \alpha + h$. Composing its coordinate maps with ϕ gives the required subspace of the original cube.

It remains to handle the case in which every restriction satisfies $a < \alpha + \varepsilon$. Let $p = \mathbb{P}(a < \alpha - \kappa)$. From (3.11) and the upper bound on a,

$$\alpha - \varepsilon \le \mathbb{E}a \le p(\alpha - \kappa) + (1-p)(\alpha + \varepsilon).$$

It follows that

$$p \le \frac{2\varepsilon}{\kappa + \varepsilon} \le \frac{2\varepsilon}{\kappa} = \frac{\delta}{4}.$$

On the other hand, if $q = \mathbb{P}(b \ge \delta/4)$, then $0 \le b \le 1$ and (3.11) give

$$\delta - \varepsilon \le \mathbb{E}b \le q + (1-q)\frac{\delta}{4}, \qquad q \ge \frac{3\delta/4 - \varepsilon}{1 - \delta/4} \ge \frac{\delta}{2}.$$

Thus there is a restriction for which

$$\nu_{3,m}(A') \ge \alpha - \kappa, \qquad \nu_{2,m}(A' \cap [2]^m) \ge \frac{\delta}{4}, \qquad A' = \phi^{-1}(A). \tag{3.12}$$

A combinatorial line in A' would map to one in A, so A' is also line-free.

Set $B = A' \cap [2]^m$ and $C_i = C_i(B)$, $C = C_1 \cap C_2$ as in Definition 3.16. Lemma 3.17 and (3.12) imply

$$\nu_{3,m}(C) \ge \frac{(\delta/4)^2}{2} = \frac{\delta^2}{32} \ge \theta.$$

Since A' is line-free, (3.8) gives

$$\nu_{3,m}(A' \cap C) \le \frac{2}{m+2} \le \frac{\delta\theta}{2} \le \frac{\alpha}{2}\nu_{3,m}(C). \tag{3.13}$$

For the next calculation write $\nu = \nu_{3,m}$ and $\beta = \alpha - \kappa$. We have $0 < \beta \le 1$. The total β-excess of A' is nonnegative by (3.12). Removing C and using (3.13), we obtain

$$\begin{aligned}\nu(A' \cap C^c) - \beta\nu(C^c) &= [\nu(A') - \beta] - [\nu(A' \cap C) - \beta\nu(C)] \\ &\ge (\alpha/2 - \kappa)\nu(C) \ge \frac{\delta\theta}{4} = 2t.\end{aligned}$$

The complement of C is the disjoint union

$$D^{(1)} = C_1^c, \qquad D^{(2)} = C_1 \cap C_2^c.$$

Each of these is an intersection of a $(1,3)$-insensitive set and a $(2,3)$-insensitive set; for $D^{(1)}$ the second set is the whole cube. Because the excesses add over this partition, one of the two sets, denoted by D, satisfies

$$\nu(A' \cap D) - \beta\nu(D) \ge t. \tag{3.14}$$

Choose a random r-coordinate restriction $\psi : [3]^r \to [3]^m$ with equal-slices background. Write $A'' = \psi^{-1}(A')$ and $D' = \psi^{-1}(D)$ for its two traces. Applying Lemma 3.15 separately to $A' \cap D$ and D, and using $\beta \le 1$, shows that the expected value of

$$\mu_{3,r}(A'' \cap D') - \beta\mu_{3,r}(D')$$

is at least $t - 2(3r/m) \ge 3t/4$. Fix a restriction for which this bound holds. The two insensitivities defining D are preserved by coordinate restriction, so D' has the required form for the tiling lemma. Since $\kappa = t/8$,

$$\mu_{3,r}(A'' \cap D') - \alpha\mu_{3,r}(D') \ge \frac{3t}{4} - \kappa\mu_{3,r}(D') \ge \frac{5t}{8} \ge \frac{t}{2}. \tag{3.15}$$

By the choice of r, there are disjoint d-dimensional subspaces $V_1, \dots, V_s$ contained in D' whose union U leaves $\mu_{3,r}(D' \setminus U) \le t/8$. Subtracting the contribution of this remainder from (3.15) gives

$$\mu_{3,r}(A'' \cap U) - \alpha\mu_{3,r}(U) \ge \frac{3t}{8}.$$

In particular U is nonempty. The left side is

$$\sum_{j=1}^{s} \mu_{3,r}(V_j) \left(\frac{|A'' \cap V_j|}{|V_j|} - \alpha \right).$$

As $\sum_j \mu_{3,r}(V_j) = \mu_{3,r}(U) \le 1$, at least one tile has relative density at least $\alpha + 3t/8$. This is at least $\alpha + h$, because $h = \delta t/128 \le 3t/8$. The image of that tile under $\phi \circ \psi$ is a d-dimensional combinatorial subspace of $[3]^n$ with the same relative density. This proves the proposition. □

Theorem 3.19 (Density Hales–Jewett for three letters)**.** *For every $0 < \delta \le 1$ there is an integer $H_3(\delta)$ such that, for every $n \ge H_3(\delta)$, each set $A \subseteq [3]^n$ with $|A| \ge \delta 3^n$ contains a combinatorial line.*

Proof. Fix δ and let $h = h(\delta) > 0$ be supplied by Proposition 3.18. Choose an integer K with $Kh > 1$, put $n_0 = 1$, and recursively choose

$$n_{j+1} = N(n_j, \delta) \qquad (0 \le j < K).$$

Set $H_3(\delta) = n_K$. First suppose $n = n_K$ and that A is line-free with density at least δ. Apply the proposition with target dimension n_{K-1}. Identifying the resulting subspace with $[3]^{n_{K-1}}$, the trace of A is still line-free and its density has increased by at least h. Apply the proposition again with target dimension n_{K-2}. Continue in this way. Composing the coordinate descriptions of combinatorial subspaces preserves their dimensions and sends their combinatorial lines to combinatorial lines of the original cube. Therefore all K applications are legitimate, and the final trace has density at least $\delta + Kh > 1$, which is impossible.

For $n > n_K$, partition $[3]^n$ into coordinate subspaces of dimension n_K by fixing the other coordinates. Their relative A-densities average to $\mu_{3,n}(A)$, so some trace has density at least δ. The case just proved gives a line in that trace, hence in A. □

4. Geometry, forms, and elementary estimates

Definition 4.1 (Triangle coordinates)**.** Fix real numbers α, β with $\beta \neq 0$ and set

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad M = \alpha I + \beta J, \qquad L_0 = 0, \quad L_1 = I, \quad L_2 = M.$$

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, $u_\theta = (\cos\theta, \sin\theta)$, and $v_\theta = Ju_\theta$. We write $\mathrm{d}\sigma(\theta) = \mathrm{d}\theta/(2\pi)$ and also use σ for its pushforward to the unit circle. The configuration space is $\mathbb{R}^2 \times \mathbb{T}$, with $p = (x, \theta)$, $\mathrm{d}\mu(p) = \mathrm{d}x\,\mathrm{d}\sigma(\theta)$, and

$$\pi_j p = x + L_j u_\theta \qquad (j = 0, 1, 2).$$

All measurable functions are represented by Borel versions when pointwise compositions are needed; equalities of functions in L^p are understood almost everywhere.

Lemma 4.2 (Similarities and normalization)**.** *Each $L_i - L_j$, $i \neq j$, is an invertible similarity. The matrix M commutes with every rotation. Every ordered noncollinear triangle is a translate of a rotation of a positive multiple of $\{0, e_1, Me_1\}$ for a matrix M of the form in Definition 4.1. For each j the map $(x, \theta) \mapsto (\pi_j(x, \theta), \theta)$ preserves μ.*

Proof. For real a, b, $(aI + bJ)^T(aI + bJ) = (a^2 + b^2)I$. Apply this to M and $M - I$, and also to I and the negatives of these matrices. The condition $\beta \neq 0$ ensures invertibility of both M and $M - I$. Every planar rotation is $\cos t\, I + \sin t\, J$, which commutes with M. Given vertices q_0, q_1, q_2, put $\ell = |q_1 - q_0| > 0$ and choose a rotation U with $Ue_1 = (q_1 - q_0)/\ell$. Write $\ell^{-1}U^{-1}(q_2 - q_0) = (\alpha, \beta)$. Noncollinearity is exactly $\beta \neq 0$, and $\{q_0, q_1, q_2\} = q_0 + \ell U\{0, e_1, Me_1\}$. The last assertion follows from translation invariance of Lebesgue measure in x, separately for each θ. □

Definition 4.3 (Counting form and its endpoint norms)**.** For a finite complex Borel measure ν on $\mathbb{R}^2$ set

$$\Lambda_\nu(f_0, f_1, f_2) = \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \prod_{j=0}^{2} f_j(x + L_j u)\,\mathrm{d}x\,\mathrm{d}\nu(u)$$

whenever the integral is absolutely convergent against $|\nu|$. For $i \in \{0, 1, 2\}$ let

$$\|\Lambda_\nu\|_i = \sup_{\|f_i\|_\infty \le 1,\ \|f_j\|_2 \le 1\ (j \neq i)} |\Lambda_\nu(f_0, f_1, f_2)|, \qquad \|\Lambda_\nu\| = \|\Lambda_\nu\|_2.$$

The total variation norm of ν is $\|\nu\|_{\mathrm{TV}} = |\nu|(\mathbb{R}^2)$. For $s > 0$, ν_s is the pushforward under $u \mapsto su$. An L^1 function is identified with the measure having that density.

Lemma 4.4 (Basic endpoint bounds and scaling)**.** *For each endpoint permutation the defining integral is absolutely convergent and*

$$|\Lambda_\nu(f_0,f_1,f_2)| \le \|\nu\|_{\mathrm{TV}}\|f_i\|_\infty \prod_{j\neq i}\|f_j\|_2, \qquad \|\Lambda_{\nu_s}\|_i = \|\Lambda_\nu\|_i.$$

The form does not depend on the choice of almost-everywhere versions of its inputs. Simultaneously translating all inputs leaves the form unchanged.

Proof. For fixed u, bound the ith factor by its essential supremum and apply Cauchy–Schwarz to the other two translated factors in x. Their L^2 norms are unchanged by translation. Integrate this bound against $|\nu|$. The same argument, or Fubini applied to a null set in any one translated variable, proves independence of versions.

For the scaling assertion put $p_i = \infty$ and $p_j = 2$ for $j \neq i$, with $1/\infty = 0$, and define $U_s^{(p)} f(x) = s^{-2/p} f(x/s)$. These are isometries on the indicated spaces. The substitutions $x = sy$ and $u = sv$ give

$$\Lambda_{\nu_s}(U_s^{(p_0)}f_0, U_s^{(p_1)}f_1, U_s^{(p_2)}f_2) = \Lambda_\nu(f_0,f_1,f_2),$$

because $\sum_j 1/p_j = 1$. Take the supremum and then apply the same argument with s^{-1}. A common translation is removed by a translation of x. □

Lemma 4.5 (Convolution as translations of the inputs)**.** *For finite measures ν, κ and endpoint inputs,*

$$\Lambda_{\nu*\kappa}(f_0,f_1,f_2) = \int \Lambda_\nu(f_0, f_1(\cdot + w), f_2(\cdot + Mw))\,\mathrm{d}\kappa(w). \tag{4.1}$$

Consequently $\|\Lambda_{\nu*\kappa}\|_i \le \|\kappa\|_{\mathrm{TV}}\|\Lambda_\nu\|_i$. More generally, any translation-invariant norm bound for the inputs of Λ_ν is inherited by $\Lambda_{\nu*\kappa}$, with the extra factor $\|\kappa\|_{\mathrm{TV}}$.

Proof. In the convolution integral replace the displacement u by $v + w$. The three vertices become x, $x + v + w$, and $x + Mv + Mw$. Fubini is justified by the absolute bound of Lemma 4.4, first against $|\nu|$ and then against $|\kappa|$. This proves the identity. Translations preserve each input norm, which proves the stated consequences whenever the corresponding absolute integral or continuous extension is defined. □

Definition 4.6 (Exponential and frequency-scale notation)**.** Write $\mathrm{e}(t) = \exp(2\pi i t)$, keeping the Fourier convention of Definition 2.1. For a function h and $R > 0$, put $h_R(u) = R^2 h(Ru)$.

Lemma 4.7 (Fourier representation)**.** *For Schwartz inputs and a finite measure ν,*

$$\Lambda_\nu(f_0,f_1,f_2) = \iint \widehat{f_0}(-\xi_1-\xi_2)\widehat{f_1}(\xi_1)\widehat{f_2}(\xi_2) \cdot \widehat{\nu}(-\xi_1 - M^T\xi_2)\,\mathrm{d}\xi_1\,\mathrm{d}\xi_2.$$

In particular, if ψ is smooth, compactly supported, and even, the form with kernel $\sigma * (\check{\psi})_R$ has multiplier $\widehat{\sigma}(b)\psi(b/R)$, where $b = \xi_1 + M^T\xi_2$.

Proof. Apply Fourier inversion from Lemma 2.5 to the last two inputs and integrate the first input in x. The resulting phase in u is $\mathrm{e}(u \cdot b)$, so its integral is $\widehat{\nu}(-b)$. Absolute integrability follows, for example, from $\|f_0\|_1\|\widehat{f_1}\|_1\|\widehat{f_2}\|_1\|\nu\|_{\mathrm{TV}} < \infty$. For the last assertion use symmetry of σ, evenness of ψ, and the convolution identity obtained directly by Fubini. The substitution $v = Ru$ in the Fourier integral gives $\widehat{h_R}(\xi) = \widehat{h}(\xi/R)$, supplying the stated scale. □

Lemma 4.8 (Density of the circle difference measure)**.** *If N is an invertible planar similarity, the pushforward of $\sigma \otimes \sigma$ under $(\theta,\varphi) \mapsto N(u_\theta - u_\varphi)$ has a nonnegative density $w_N \in L^1(\mathbb{R}^2)$ with integral one.*

Proof. The absolute Jacobian of this map is $|\det N|\,|\det(v_\theta, v_\varphi)| = |\det N|\,|\sin(\varphi-\theta)|$. It is nonzero except on the two one-dimensional sets $\varphi = \theta$ and $\varphi = \theta + \pi$ modulo 2π, which have zero product measure. The complement has a countable cover by open sets on which the map is a diffeomorphism onto its image, by the inverse function theorem and second countability. Partition this complement into measurable pieces subordinate to that cover. On each piece the change-of-variables theorem shows that the pushforward is absolutely continuous. Their countable sum is absolutely continuous as well, and the omitted set has zero measure. The Radon–Nikodym density is nonnegative and has integral equal to the original mass, namely one. □

Lemma 4.9 (Weighted circle compactness and small-set control)**.** *Let N be an invertible planar similarity and W a measurable function with $|W| \le 1$. Define*

$$Tf(z) = \int W(z,\theta) f(z + Nu_\theta)\,\mathrm{d}\sigma(\theta).$$

There is a nondecreasing function $\omega_N : [0,\infty) \to [0,1]$, with $\omega_N(s) \to 0$ as $s \downarrow 0$, independent of W, such that

$$\|\sqrt{e}\,T\|_{L^2\to L^2} \le \omega_N(\|e\|_1) \quad \text{if } 0 \le e \le 1,\ e \in L^1.$$

If $W(z,\theta) = 0$ for z outside a bounded set, then T is compact on $L^2(\mathbb{R}^2)$.

Proof. Cauchy–Schwarz in the probability measure σ and translation invariance give $\|T\| \le 1$. Its adjoint is

$$T^*g(y) = \int \overline{W(y - Nu_\varphi, \varphi)}\, g(y - Nu_\varphi)\,\mathrm{d}\sigma(\varphi).$$

Consequently

$$TT^*g(z) = \iint W(z,\theta)\overline{W(z + Nu_\theta - Nu_\varphi, \varphi)} \cdot g(z + Nu_\theta - Nu_\varphi)\,\mathrm{d}\sigma(\theta)\,\mathrm{d}\sigma(\varphi).$$

By Lemma 4.8 this is an integral operator with a measurable kernel $K(z,y)$ satisfying $|K(z,y)| \le w_N(y-z)$. One precise way to obtain the kernel is to push forward the weighted measure in (z,θ,φ) under $(z,\theta,\varphi) \mapsto (z, z + Nu_\theta - Nu_\varphi)$; its total variation is bounded by $w_N(y-z)\,\mathrm{d}z\,\mathrm{d}y$, so Radon–Nikodym applies on bounded z sets and then consistently on all of $\mathbb{R}^2$. The resulting identity first holds in pairings with bounded compactly supported tests. The domination by w_N and Lemma 2.2 extend it to L^2 by density.

For $H > 0$ split the kernel according to $w_N(y-z) \le H$ and $w_N(y-z) > H$. In the first part the kernel of $\sqrt{e}\,TT^*\sqrt{e}$ is bounded in modulus by $H\sqrt{e(z)e(y)}$, a positive rank-one kernel of operator norm $H\|e\|_1$. In the second part, Young's inequality and $\sqrt{e} \le 1$ bound the operator norm by $\|w_N \mathbf{1}_{\{w_N > H\}}\|_1$. Thus

$$\|\sqrt{e}\,T\|^2 \le H\|e\|_1 + \|w_N \mathbf{1}_{\{w_N > H\}}\|_1. \tag{4.2}$$

Take the minimum with one and then the infimum over $H > 0$ of the square root of the right side. The resulting modulus is nondecreasing. It tends to zero because the second term can first be made small by increasing H, and the first can then be made small by decreasing $\|e\|_1$.

If W has bounded z support X, both variables of K are in X. The part with $w_N \le H$ is a bounded kernel on $X \times X$, hence is compact by Lemma 2.10. The remaining part has norm tending to zero as $H \to \infty$. The same lemma makes their operator-norm limit TT^* compact. Lemma 2.11 now gives compactness of T. □

Lemma 4.10 (The trilinear L^2 estimate)**.** *There is a constant C_M such that*

$$|\Lambda_\sigma(f_0, f_1, f_2)| \le C_M \prod_{j=0}^{2} \|f_j\|_2. \tag{4.3}$$

The integral is absolutely convergent for these inputs. For every finite measure κ, the same assertions hold for $\sigma * \kappa$ with the additional factor $\|\kappa\|_{\mathrm{TV}}$.

Proof. Consider $F(\theta, \varphi) = Mu_\varphi - u_\theta$. At (θ_0, θ_0) its derivative has columns $-v_{\theta_0}$ and Mv_{θ_0}, whose determinant has absolute value $|\beta| > 0$. There is a sufficiently small interval I about each θ_0 on which F is one-to-one on $I \times I$ and its absolute Jacobian is bounded below. For completeness, choose I so that DF differs from the invertible matrix $DF(\theta_0, \theta_0)$ by less than half its least singular value. Integrating this difference along segments in $I \times I$ proves injectivity and the lower Jacobian bound. Compactness of the circle supplies a finite partition into intervals with these properties.

For such an interval let $B_I(f_1, f_2)(x) = \int_I f_1(x + u_\theta) f_2(x + Mu_\theta)\, \mathrm{d}\sigma(\theta)$. Initially take bounded compactly supported inputs. Expand $\|B_I\|_2^2$, take absolute values, and apply Cauchy–Schwarz on (x, θ, φ) to the factors

$$|f_1(x + u_\theta) f_2(x + Mu_\varphi)|, \qquad |f_1(x + u_\varphi) f_2(x + Mu_\theta)|.$$

The two resulting square integrals are identical by interchanging θ and φ. It follows that

$$\begin{aligned} \|B_I(f_1, f_2)\|_2^2 &\le \int_{I^2} \int |f_1(x + u_\theta)|^2 |f_2(x + Mu_\varphi)|^2 \, \mathrm{d}x \, \mathrm{d}\sigma(\theta) \, \mathrm{d}\sigma(\varphi) \\ &= \int |f_1(y)|^2 \int_{I^2} |f_2(y + F(\theta, \varphi))|^2 \, \mathrm{d}\sigma(\theta) \, \mathrm{d}\sigma(\varphi) \, \mathrm{d}y \\ &\le C_M \|f_1\|_2^2 \|f_2\|_2^2. \end{aligned}$$

The last inequality is the change of variables through F. Sum the L^2 norms over the finite partition and pair with f_0 by Cauchy–Schwarz. Applying the same bound to absolute values, followed by monotone approximation, proves both absolute convergence and the estimate for arbitrary L^2 inputs. Finally use the identity and translation argument in Lemma 4.5, first for absolute values against $|\kappa|$. □

Lemma 4.11 (Small support in an L^2 input)**.** *There is a nondecreasing modulus $\omega_M(s) \to 0$ as $s \downarrow 0$ with the following property. Suppose i, j, k are distinct, $f_i \in L^\infty$, $f_j, f_k \in L^2$, and f_j is supported on a measurable set of area at most s. Then, for every finite measure κ,*

$$|\Lambda_{\sigma * \kappa}(f_0, f_1, f_2)| \le \|\kappa\|_{\mathrm{TV}} \omega_M(s) \|f_i\|_\infty \|f_j\|_2 \|f_k\|_2.$$

The assertion for σ is included by taking $\kappa = \delta_0$.

Proof. For σ, bound the ith input by its supremum and set $z = x + L_j u_\theta$. The remaining integral is bounded by

$$\|f_i\|_\infty \int |f_j(z)| \left(\int |f_k(z + (L_k - L_j)u_\theta)| \, \mathrm{d}\sigma(\theta) \right) \mathrm{d}z.$$

Since f_j is supported on a set A of measure at most s, Cauchy–Schwarz and Lemma 4.9, with $e = \mathbf{1}_A$, bound this by $\omega_{L_k - L_j}(s) \|f_i\|_\infty \|f_j\|_2 \|f_k\|_2$. Take the maximum over the six possible edge maps, which are invertible similarities by Lemma 4.2. For a convolution by κ, translate the inputs as in Lemma 4.5; the small support is merely translated and its area is unchanged. Integrate the bound against $|\kappa|$. □

Remark 4.12. The symbol ω_M may be increased by a constant depending only on M whenever finitely many of the edge moduli are combined. It will always be nondecreasing and tend to zero at zero. All constants denoted by C_M may change from one inequality to the next. These conventions do not assert any quantitative rate of decay.

5. Finite-dimensional scalar kernels

Definition 5.1 (Degrees and cores)**.** Let $X_0, X_1, X_2 \subset \mathbb{R}^2$ be bounded measurable sets, and let $E \subset \mathbb{R}^2 \times \mathbb{T}$ be measurable, with $\pi_j(p) \in X_j$ for almost every $p \in E$ and every j. The degree of $z \in \mathbb{R}^2$ at vertex j is

$$d_{j,E}(z) = \int_{\mathbb{T}} \mathbf{1}_E(z - L_j u_\theta, \theta)\, \mathrm{d}\sigma(\theta).$$

The data $(E; X_0, X_1, X_2)$ form a *core of degree* c if $\mu(E) > 0$ and $d_{j,E} \geq c > 0$ almost everywhere on X_j, for every j. Sets and their defining relations may throughout be modified on null sets.

Lemma 5.2 (Integration along a vertex)**.** *For every nonnegative measurable function g on $\mathbb{R}^2$,*

$$\int_E g(\pi_j p)\, \mathrm{d}\mu(p) = \int_{\mathbb{R}^2} g(z) d_{j,E}(z)\, \mathrm{d}z.$$

In particular, $0 \leq d_{j,E} \leq 1$, $\int d_{j,E} = \mu(E)$, and pullback by π_j sends planar null sets to μ-null sets.

Proof. Apply Tonelli's theorem, and at fixed θ make the translation $z = x + L_j u_\theta$. Its Jacobian is one. The bound on the degree follows because σ is a probability measure. Taking $g = 1$ gives its integral, and taking g to be the indicator of a null set gives the last assertion. □

Lemma 5.3 (Extraction of a positive-degree core)**.** *If $\mu(E) > 0$, then for every*

$$0 < c < \frac{\mu(E)}{\sum_{j=0}^{2} |X_j|}$$

there are measurable $Y_j \subset X_j$ and $F \subset E$ such that $(F; Y_0, Y_1, Y_2)$ is a core of degree c. Moreover,

$$\mu(E \setminus F) \leq c \sum_{j=0}^{2} |X_j|.$$

Proof. Set $E^{(0)} = E$ and $X_j^{(0)} = X_j$. Having defined these objects at step m, delete

$$V_j^{(m)} = \{z \in X_j^{(m)} : d_{j,E^{(m)}}(z) < c\}, \qquad X_j^{(m+1)} = X_j^{(m)} \setminus V_j^{(m)},$$

and let

$$E^{(m+1)} = E^{(m)} \cap \bigcap_{j=0}^{2} \pi_j^{-1}(X_j^{(m+1)}).$$

By Lemma 5.2 and the union bound,

$$\mu(E^{(m)} \setminus E^{(m+1)}) \leq \sum_j \int_{V_j^{(m)}} d_{j,E^{(m)}}(z)\, \mathrm{d}z \leq c \sum_j |V_j^{(m)}|.$$

For each j, the deleted vertex sets are disjoint as m varies. Consequently the sets $Y_j = \bigcap_m X_j^{(m)}$ and $F = \bigcap_m E^{(m)}$ satisfy the stated loss bound. Its right-hand side is smaller than $\mu(E)$, so $\mu(F) > 0$. For almost every $z \in Y_j$, survival at every step gives $d_{j,E^{(m)}}(z) \geq c$ for every m. Continuity of the integral under decreasing bounded integrands gives $d_{j,F}(z) = \lim_m d_{j,E^{(m)}}(z) \geq c$. □

Definition 5.4 (The scalar incidence operator)**.** On a core $(E; X_0, X_1, X_2)$ let

$$\mathcal{H}_E = \bigoplus_{j=0}^{2} L^2(X_j; \mathbb{R}), \qquad D_E : \mathcal{H}_E \longrightarrow L^2(E, \mu), \qquad D_E f = \sum_{j=0}^{2} f_j \circ \pi_j, \qquad K_E = \ker D_E.$$

The norm on $\mathcal{H}_E$ is the square root of the sum of the three squared L^2 norms. Each component is extended by zero when regarded as a function on all of $\mathbb{R}^2$.

Lemma 5.5 (Finite-dimensionality and a distance estimate)**.** *For every core E, the space K_E is finite dimensional. There is $C_E < \infty$ such that*

$$\operatorname{dist}_{\mathcal{H}_E}(f, K_E) \le C_E \, \|D_E f\|_{L^2(E,\mu)} \qquad (f \in \mathcal{H}_E).$$

The same constant is valid for every translate of E in the base variable.

Proof. The inequality $|a_0 + a_1 + a_2|^2 \le 3 \sum_j |a_j|^2$, followed by Lemma 5.2, shows that $\|D_E\| \le \sqrt{3}$. That lemma also identifies the blocks of $D_E^* D_E$. The diagonal block at vertex j is multiplication by $d_{j,E}$. For $i \ne j$ the block is

$$T_{ji} f_i(z) = \int_{\mathbb{T}} \mathbf{1}_E(z - L_j u_\theta, \theta) f_i(z + (L_i - L_j) u_\theta) \, d\sigma(\theta), \qquad z \in X_j.$$

Its weight has absolute value at most one and bounded support in z. Each $L_i - L_j$ is a nonzero similarity, so Lemma 4.9 makes every off-diagonal block compact.

Write $D_E^* D_E = A + K$, where A is the diagonal multiplication operator and K is compact and self-adjoint. If E has degree c, then $cI \le A \le I$. Thus

$$B = A^{-1/2} D_E^* D_E A^{-1/2} = I + A^{-1/2} K A^{-1/2}$$

is a nonnegative compact perturbation of the identity. Lemma 2.12 implies that $\ker B$ is finite dimensional and that some $\gamma > 0$ satisfies

$$\langle Bg, g \rangle \ge \gamma \operatorname{dist}(g, \ker B)^2.$$

Here $\ker B = A^{1/2} K_E$. Substituting $g = A^{1/2} f$ and using $\left\| A^{1/2}(f - k) \right\| \ge \sqrt{c} \, \|f - k\|$ for $k \in K_E$ gives

$$\|D_E f\|_2^2 \ge \gamma c \operatorname{dist}(f, K_E)^2.$$

It also proves finite-dimensionality of K_E. Base translation induces unitary identifications of both Hilbert spaces and intertwines the incidence operators, so it preserves this estimate. □

Lemma 5.6 (Uniform integrability of kernel elements)**.** *There is a nondecreasing function $\omega : [0, \infty) \to [0, \infty)$, depending only on M, with $\omega(s) \to 0$ as $s \downarrow 0$, such that the following holds. On any core G of degree $t > 0$, if $f \in K_G$, $0 \le e \le 1$, and $e \in L^1(\mathbb{R}^2)$, then*

$$\left\| \sqrt{e} \, f_j \right\|_2 \le C_M t^{-1} \omega(\|e\|_1) \, \|f\|_{\mathcal{H}_G} \qquad (j = 0, 1, 2).$$

The modulus is independent of the core and of its vertex sets.

Proof. Take ω to be the maximum of the finitely many moduli supplied by Lemma 4.9 for $L_i - L_j$, $i \ne j$. The jth block of $D_G^* D_G f = 0$ reads

$$d_{j,G} f_j = - \sum_{i \ne j} T_{ji} f_i.$$

Multiplication by $\sqrt{e}$, the lower bound $d_{j,G} \ge t$ on the support of f_j, and that lemma give

$$t \left\| \sqrt{e} \, f_j \right\|_2 \le \sum_{i \ne j} \left\| \sqrt{e} \, T_{ji} f_i \right\|_2 \le \omega(\|e\|_1) \sum_{i \ne j} \|f_i\|_2 .$$

The last sum is bounded by $\sqrt{2} \, \|f\|_{\mathcal{H}_G}$. □

Lemma 5.7 (Stability under removal of a small set)**.** *Fix a core $(E; X_0, X_1, X_2)$ of degree c and a number $t > 0$. There is $\eta_0 > 0$ such that, whenever $(F; Y_0, Y_1, Y_2)$ is a core of degree t, with $F \subset E$, $Y_j \subset X_j$, and $\mu(E \setminus F) < \eta_0$, the restriction map*

$$R : K_E \longrightarrow K_F, \qquad Rf = (f_j|_{Y_j})_{j=0}^2,$$

is an isomorphism. The threshold may be chosen using only c, t, C_E, and the fixed modulus in Lemma 5.6; in particular, the same threshold works after a base translation.

Proof. Set $\eta = \mu(E \setminus F)$. Restriction takes K_E to K_F because $F \subset E$. Every incidence of a vertex in $X_j \setminus Y_j$ is absent from F. Therefore

$$c|X_j \setminus Y_j| \le \int_{X_j \setminus Y_j} d_{j,E}(z)\,\mathrm{d}z \le \eta.$$

Applying Lemma 5.6 on E, with $e = \mathbf{1}_{X_j \setminus Y_j}$, proves

$$\sum_j \left\| f_j \mathbf{1}_{X_j \setminus Y_j} \right\|_2^2 \le C_M c^{-2} \omega(\eta/c)^2 \, \|f\|_{\mathcal{H}_E}^2 \qquad (f \in K_E).$$

For sufficiently small η, this makes R injective.

For the reverse dimension inequality, extend $f \in K_F$ by zero to $\widetilde{f} \in \mathcal{H}_E$. Define $e_j = d_{j,E\setminus F}$, so $0 \le e_j \le 1$ and $\int e_j = \eta$. Since $D_E \widetilde{f} = 0$ almost everywhere on F,

$$\begin{aligned} \left\| D_E \widetilde{f} \right\|_2^2 &\le 3 \sum_j \int_{E\setminus F} |\widetilde{f}_j(\pi_j p)|^2 \,\mathrm{d}\mu(p) \\ &= 3 \sum_j \left\| \sqrt{e_j}\, \widetilde{f}_j \right\|_2^2 \le C_M t^{-2} \omega(\eta)^2 \, \|f\|_{\mathcal{H}_F}^2 . \end{aligned}$$

Lemma 5.5 now gives

$$\operatorname{dist}(\widetilde{f}, K_E) \le C_M C_E t^{-1} \omega(\eta) \, \|f\|_{\mathcal{H}_F} .$$

Choose η_0 so that the coefficient is less than one. The orthogonal projection of the zero extension onto K_E is then injective on K_F: if its value were zero, the displayed inequality would force $f = 0$. Both kernels are finite dimensional by Lemma 5.5. The two injections imply equality of their dimensions, and the already injective restriction is consequently surjective. The stated dependence of the threshold follows from the estimates just proved. □

6. Analytic pieces and measurable rigidity

Lemma 6.1 (A common core for nearby translates)**.** *Let E be a core of degree c, and define*

$$E_h = \{(x,\theta) : (x+h,\theta) \in E\}, \qquad \eta(h) = \mu(E \setminus E_h).$$

For all sufficiently small $h \in \mathbb{R}^2$, there is a core $(F_h; Y_{0,h}, Y_{1,h}, Y_{2,h})$ of degree $c/4$, with

$$F_h \subset E \cap E_h, \qquad Y_{j,h} \subset X_j \cap (X_j - h),$$

such that

$$\mu(E \setminus F_h) = \mu(E_h \setminus F_h) \le 4\eta(h), \qquad |X_j \setminus Y_{j,h}| \longrightarrow 0 \quad (h \to 0).$$

Proof. Translation continuity in $L^1(\mathbb{R}^2 \times \mathbb{T})$ gives $\mu(E \triangle E_h) \to 0$; in particular $\eta(h) \to 0$. Start with $E \cap E_h$ on the original vertex sets X_j, and perform the deletion process from Lemma 5.3 at threshold $t = c/4$. Write V_j for the union of the deleted vertices and $V = \sum_j |V_j|$. If L denotes the measure of further incidences removed after the initial removal of $E \setminus E_h$, the same accounting as in that lemma gives $L \le tV$. On the other hand, the degrees in the original core yield

$$cV \le \sum_j \int_{V_j} d_{j,E}(z)\,\mathrm{d}z \le 3\mu(E \setminus F_h) = 3(\eta(h) + L).$$

Every removed configuration is counted at most three times in this sum. Thus $cV \le 3\eta(h) + 3cV/4$, so $V \le 12\eta(h)/c$ and $\mu(E \setminus F_h) \le \eta(h) + (c/4)V \le 4\eta(h)$. For small h this is less than $\mu(E)$, and the surviving set is a positive-measure core. Surviving vertices have positive degree in $E \cap E_h$, so their sets may be taken inside $X_j \cap (X_j - h)$ up to null sets. Since $\mu(E_h) = \mu(E)$, the two displayed configuration losses are equal. The bound on V proves the convergence of the vertex losses. □

Lemma 6.2 (Translation matrices)**.** *Let $\phi^1, \dots, \phi^d$ be an orthonormal basis of K_E, and write*

$$\Phi_j(z) = (\phi_j^1(z), \dots, \phi_j^d(z)),$$

with the components extended by zero. For sufficiently small h there are sets $Y_{j,h}$ as in Lemma 6.1 and matrices $C(h) \in \mathrm{GL}_d(\mathbb{R})$ such that

$$\Phi_j(z+h) = \Phi_j(z)C(h) \quad \text{for almost every } z \in Y_{j,h}.$$

On a sufficiently small neighborhood of zero these matrices form a continuous local homomorphism:

$$C(0) = I, \qquad C(h+k) = C(h)C(k)$$

whenever $h, k, h+k$ lie in that neighborhood.

Proof. The case $d = 0$ is vacuous, so assume $d \geq 1$. The translated rows $\Phi_j(\cdot + h)$ give an orthonormal basis for K_{E_h}. By Lemmas 6.1 and 5.7, restriction identifies both K_E and K_{E_h} with K_{F_h} for all sufficiently small h. The threshold for E_h is the same as for E. Two bases of the same finite-dimensional space differ by a unique invertible matrix, which defines $C(h)$ and gives the asserted row identities.

Orthonormality says

$$\sum_j \int_{X_j} \Phi_j(z)^{\mathsf{T}} \Phi_j(z)\, \mathrm{d}z = I.$$

Consequently $A_h = \sum_j \int_{Y_{j,h}} \Phi_j^{\mathsf{T}} \Phi_j \to I$, by absolute continuity of the integrals and the vertex-loss estimate. Multiplying the row identity on the left by $\Phi_j(z)^{\mathsf{T}}$ and integrating gives

$$A_h C(h) = \sum_j \int_{Y_{j,h}} \Phi_j(z)^{\mathsf{T}} \Phi_j(z+h)\, \mathrm{d}z.$$

The right-hand side minus A_h tends to zero by Cauchy–Schwarz and L^2 translation continuity of each row. Hence $C(h) \to I$ as $h \to 0$.

To verify composition, set

$$Z_j = Y_{j,h} \cap (Y_{j,k} - h) \cap Y_{j,h+k}.$$

For almost every $z \in Z_j$, the three row identities give

$$\Phi_j(z)C(h+k) = \Phi_j(z+h+k) = \Phi_j(z+h)C(k) = \Phi_j(z)C(h)C(k).$$

The complements of Z_j in X_j have measure tending to zero as $h, k \to 0$. Indeed, besides the three vertex losses, it is enough to allow for $|X_j \triangle (X_j - h)|$, which tends to zero by translation continuity of the indicator of X_j. Thus the Gram matrix $\sum_j \int_{Z_j} \Phi_j^{\mathsf{T}} \Phi_j$ is invertible for all sufficiently small h, k. Multiplying and integrating the last row identity yields $C(h+k) = C(h)C(k)$. In particular $C(0) = I$ and $C(-h) = C(h)^{-1}$ where defined. Continuity at any nearby h_0 follows from continuity at zero and $C(h_0 + h) = C(h_0)C(h)$. □

Proposition 6.3 (Analytic pieces for a scalar kernel)**.** *Every positive-measure core E has a measurable subset $E' \subset E$ of positive measure such that each component of an orthonormal basis of K_E, evaluated at its vertex, agrees on E' with a real analytic function defined on all of $\mathbb{R}^2$.*

Proof. If $K_E = \{0\}$, the conclusion is vacuous and we take $E' = E$. Otherwise apply Lemma 2.21 to the local homomorphism of Lemma 6.2. It gives commuting matrices T_1, T_2 and an everywhere real analytic function

$$P(h) = \exp(h_1 T_1 + h_2 T_2)$$

which equals $C(h)$ near zero. Use the rows Φ_j from Lemma 6.2, taking finite Borel representatives for them, and write $z_j = \pi_j p$. The translation identity implies

$$\Phi_j(z_j) = \Phi_j(z_j - h_j)P(h_j)$$

whenever $z_j \in Y_{j,h_j} + h_j$, apart from a planar null set. Lemma 5.2 shows that the exceptional configurations in E have measure at most

$$|X_j \setminus (Y_{j,h_j} + h_j)| \le |X_j \triangle (X_j + h_j)| + |X_j \setminus Y_{j,h_j}|,$$

which tends to zero as $h_j \to 0$. Choose a ball B centered at zero so small that, for every triple $(h_0, h_1, h_2) \in B^3$, all three row identities hold simultaneously on a subset of E of measure at least $\mu(E)/2$.

The set of (p, h_0, h_1, h_2) satisfying these identities is jointly measurable: this uses the Borel representatives and the explicit continuous function P, not any measurable choice of the cores. Its measure in $E \times B^3$ is at least $\mu(E)|B|^3/2 > 0$. For each fixed p make the change of variables $b_j = z_j - h_j$, $j = 0, 1, 2$, in the six translation variables. The Jacobian has absolute value one. Fubini's theorem gives a fixed triple (b_0, b_1, b_2) for which the set of configurations satisfying

$$\Phi_j(z_j) = \Phi_j(b_j) P(z_j - b_j), \qquad j = 0, 1, 2,$$

has positive μ-measure. Take this set to be E'. For each j, the row-valued function $z \mapsto \Phi_j(b_j) P(z - b_j)$ is real analytic on all of $\mathbb{R}^2$. Its components supply the required analytic representatives. □

Lemma 6.4 (Rigidity for analytic fields)**.** *Let $a_j : \mathbb{R}^2 \to \mathbb{R}^2$ be real analytic and suppose that, for every $(x, \theta) \in \mathbb{R}^2 \times \mathbb{T}$,*

$$\sum_{j=0}^{2} a_j(x + L_j u_\theta) = 0, \qquad v_\theta \cdot \big(a_1(x + u_\theta) + M^{\mathsf{T}} a_2(x + M u_\theta)\big) = 0.$$

Then

$$a_1(x + u_\theta) + M^{\mathsf{T}} a_2(x + M u_\theta) = 0 \qquad \text{for every } (x, \theta).$$

Proof. Fix the notation

$$A = a_0(x), \quad B = a_1(x + u_\theta), \quad C = a_2(x + M u_\theta), \quad u = u_\theta, \quad v = Ju.$$

Since u, v are an orthonormal basis, the second hypothesis says that $B + M^{\mathsf{T}} C = tu$ for a smooth real scalar function $t = t(x, \theta)$. Combining this with $A + B + C = 0$ gives

$$(M^{\mathsf{T}} - I)C = A + tu, \qquad -(M^{\mathsf{T}} - I)B = M^{\mathsf{T}} A + tu. \tag{6.1}$$

Put $K = Da_0(x)$, with column-vector convention for the derivative, and define

$$X_1 = \partial_\theta - v \cdot \nabla_x, \qquad X_2 = \partial_\theta - Mv \cdot \nabla_x.$$

The chain rule gives $X_1 B = 0$ and $X_2 C = 0$. Applying X_1 to the second identity in (6.1), and X_2 to the first, yields

$$(X_1 t)u + tv = M^{\mathsf{T}} K v, \qquad (X_2 t)u + tv = KMv. \tag{6.2}$$

Take the inner product of the second identity with v. Since $u \cdot v = 0$ and $|v| = 1$, it gives

$$t(x, \theta) = v^{\mathsf{T}} K M v.$$

In particular $t(x, \theta + \pi) = t(x, \theta)$. In the expanded versions of (6.2), the two spatial transport terms $(v \cdot \nabla_x t)u$ and $(Mv \cdot \nabla_x t)u$ are unchanged by $\theta \mapsto \theta + \pi$. The terms $\partial_\theta(tu)$, $M^{\mathsf{T}} K v$, and KMv change sign. Subtracting each equation at $\theta + \pi$ from the equation at θ therefore gives

$$\partial_\theta(tu) = M^{\mathsf{T}} K v = KMv.$$

As v ranges over the unit circle, this proves $M^{\mathsf{T}} K = KM$. Integration in θ gives $tu = KMu + c(x)$. Both tu and KMu change sign under $\theta \mapsto \theta + \pi$, so $c(x) = 0$.

Thus KM sends every line through the origin into itself. A real linear map on $\mathbb{R}^2$ with this property is scalar: applying it to $e_1, e_2, e_1 + e_2$ shows that its two diagonal eigenvalues agree. Hence

$KM = \lambda(x)I$. Since M is invertible, K commutes with M. The identity $M^{\mathsf{T}}K = KM$ consequently becomes

$$(M^{\mathsf{T}} - M)K = 0.$$

But $M^{\mathsf{T}} - M = -2\beta J$ is invertible because $\beta \neq 0$. It follows that $K = 0$, and then $t = 0$. Therefore $B + M^{\mathsf{T}}C = tu = 0$, as required. □

Theorem 6.5 (Measurable stationarity forces zero displacement frequency)**.** *Let $(E; X_0, X_1, X_2)$ be a core of positive degree, and let $a_j \in L^2(X_j; \mathbb{R}^2)$ satisfy, for almost every $p = (x, \theta) \in E$,*

$$\sum_{j=0}^{2} a_j(\pi_j p) = 0, \qquad v_\theta \cdot (a_1(\pi_1 p) + M^{\mathsf{T}} a_2(\pi_2 p)) = 0.$$

Then

$$a_1(\pi_1 p) + M^{\mathsf{T}} a_2(\pi_2 p) = 0 \qquad \text{for almost every } p \in E.$$

Proof. Suppose instead that the asserted vector is nonzero on a measurable set $G \subset E$ of positive measure. By Lemma 5.3, G contains a positive-measure core F on bounded vertex subsets. Restrict the fields to these subsets. Each of the two scalar-coordinate triples of the fields belongs to K_F by the first stationary equation. Express these triples in an orthonormal basis of K_F and apply Proposition 6.3. On a subset $F' \subset F$ of positive measure, the fields at their vertices agree with real analytic fields $\widetilde{a}_j : \mathbb{R}^2 \to \mathbb{R}^2$.

The two stationary expressions for these analytic fields are real analytic on the connected real analytic manifold $\mathbb{R}^2 \times \mathbb{T}$. They vanish on F', except on a null set. Lemma 2.22, applied to each scalar component on $\mathbb{R}^2 \times \mathbb{T}$, shows that both stationary equations hold everywhere for the fields $\widetilde{a}_j$. Lemma 6.4 makes their displacement-frequency vector identically zero. On F' it equals the displacement-frequency vector of the original fields, which was nonzero almost everywhere because $F' \subset G$. This contradiction proves the theorem. □

7. Translation correlations and local cancellation

Definition 7.1 (Admissible sequences and translates)**.** Fix a bounded box $Q \subset \mathbb{R}^2$ and a constant $C < \infty$. An admissible sequence consists of $R_n \to \infty$ and complex-valued functions $f_{j,n}$, $j \in \{0, 1, 2\}$, such that

$$|f_{j,n}| \leq \mathbf{1}_Q, \qquad \operatorname{Lip}(f_{j,n}) \leq C R_n.$$

For $s \in \mathbb{R}^2$ write

$$T^s_{j,n}(z) = f_{j,n}(z + s/R_n).$$

Weak-* convergence on a vertex space or on configuration space means convergence against every L^1 test function on that space.

Lemma 7.2 (Simultaneous correlation limits)**.** *Every admissible sequence has a subsequence with the following properties. For all real shifts $s, t, s_0, s_1, s_2 \in \mathbb{R}^2$, the limits*

$$T^s_{j,n}\overline{T^t_{j,n}} \overset{*}{\rightharpoonup} k_j(z; s, t),$$

$$\prod_{j=0}^{2} T^{s_j}_{j,n}(\pi_j p) \overset{*}{\rightharpoonup} h(p; s_0, s_1, s_2)$$

exist. The limits have representatives that are continuous in all shift variables off a fixed null set in the respective underlying space. They are bounded by one, and their Lipschitz constants in the shifts are bounded in terms of C. The three-point convergence also holds when the shifts s_j are replaced by bounded measurable functions of p.

Proof. First take all shifts in $\mathbb{Q}^2$. There are countably many two-point and three-point products. Each is bounded by one in L^∞. Lemma 2.13 gives all these limits on one subsequence by its countable diagonal construction. For example,

$$|T^s_{j,n}(z) - T^t_{j,n}(z)| \le C|s-t|.$$

The elementary product estimate therefore gives uniform Lipschitz bounds for every correlation as a function of its shifts. These inequalities pass to the weak-* limits. Since there are only countably many rational shifts, they hold simultaneously outside one null set. Complete the rational shift variables by continuity at every point outside that set. The same inequalities, tested against an arbitrary L^1 function, show that the original products converge weak-* at every real choice of shifts.

The resulting limit functions are jointly measurable: they are measurable in the underlying variable at rational shifts and are continuous in the shift variables. For bounded measurable shifts, round each coordinate to a rational grid of mesh ε. The rounded shifts take finitely many values. Convergence for them follows by testing on their measurable level sets. Both the original products and their limits change by at most $O_C(\varepsilon)$ when the shifts are rounded. Letting $\varepsilon \downarrow 0$ proves the last assertion. □

Lemma 7.3 (Measurable spectral measures)**.** *For a subsequence as in Lemma 7.2, there are measurable kernels of finite positive Borel measures $z \mapsto \nu_{j,z}$ on $\mathbb{R}^2$, of mass at most one, such that*

$$k_j(z;s,t) = \int_{\mathbb{R}^2} \mathrm{e}((s-t)\cdot\xi)\, \mathrm{d}\nu_{j,z}(\xi)$$

for almost every z and all $s,t \in \mathbb{R}^2$.

Proof. For fixed s,t, changing variables by $z \mapsto z + t/R_n$ in a compactly supported test function shows that

$$k_j(z;s,t) = k_j(z;s-t,0)$$

almost everywhere. Translation continuity of the test in L^1 justifies this assertion. Take rational shifts first and then use Lemma 7.2 to obtain one exceptional null set for all real shifts. Write $c_{j,z}(s) = k_j(z;s,0)$.

For any finite family of rational shifts s_a and rational complex coefficients c_a, the nonnegative functions

$$\left|\sum_a c_a T^{s_a}_{j,n}(z)\right|^2$$

have weak-* limit $\sum_{a,b} c_a \overline{c_b} c_{j,z}(s_a - s_b)$. It is nonnegative almost everywhere. Countability and continuity extend this conclusion to all shifts and coefficients on one common full-measure set. Thus $c_{j,z}$ is a Lipschitz positive-definite function there, by the shift bound in Lemma 7.2. Theorem 2.17 supplies a finite positive measure $\nu_{j,z}$ with the asserted Fourier representation. Its mass is $c_{j,z}(0)$, which lies in $[0,1]$. Set $\nu_{j,z} = 0$ on the exceptional set.

For each shift s, $z \mapsto c_{j,z}(s)$ is measurable. Set it equal to zero on the same exceptional set. Lemma 2.18 now proves that $z \mapsto \nu_{j,z}(B)$ is measurable for every Borel set B, which is the kernel assertion. □

Definition 7.4 (Polynomial translates)**.** Write $e_s(\xi) = \mathrm{e}(s\cdot\xi)$. A trigonometric polynomial on $\mathbb{R}^2$ is a finite complex linear combination of the functions e_s. For $a = \sum_s c_s e_s$ put

$$F_{j,n}[a](z) = \sum_s c_s T^s_{j,n}(z).$$

For a finite sum of elementary tensors $G = \sum_{d=1}^D b_d \otimes c_d$, with b_d, c_d trigonometric polynomials, put

$$G_n(p) = \sum_{d=1}^D F_{1,n}[b_d](\pi_1 p) F_{2,n}[c_d](\pi_2 p).$$

When p is fixed, abbreviate $\nu_j = \nu_{j,\pi_j p}$.

Lemma 7.5 (Factorization of two-vertex quadratic correlations)**.** *For the notation of Definition 7.4,*

$$|F_{0,n}[a](\pi_0 p)|^2 \overset{*}{\rightharpoonup} \int |a|^2 \, \mathrm{d}\nu_0,$$

$$|G_n(p)|^2 \overset{*}{\rightharpoonup} \int |G|^2 \, \mathrm{d}(\nu_1 \otimes \nu_2).$$

Proof. The first assertion follows by expanding the square and using Lemma 7.3. Pullback by $\pi_0(x,\theta) = x$ preserves the required convergence, by integration of the test in θ.

For the second assertion, first consider a term in the expansion of $|G_n|^2$. It has the form

$$A_n(\pi_1 p) B_n(\pi_2 p),$$

where A_n and B_n are products of two polynomial translates. Their one-vertex weak-* limits A and B are the corresponding integrals against $\nu_{1,z}$ and $\nu_{2,z}$. Take a bounded compactly supported measurable test $\varphi(p)$. In coordinates $z = x + u_\theta$, its pairing with the displayed product is

$$\int A_n(z)\,(T_\varphi B_n)(z) \, \mathrm{d}z, \qquad T_\varphi b(z) = \int \varphi(z - u_\theta, \theta) b(z + (M - I)u_\theta) \, \mathrm{d}\sigma(\theta).$$

The weight of T_φ is bounded and has bounded support in z. Since $M - I$ is a nonzero similarity, Lemma 4.9 implies that T_φ is compact on L^2. Only fixed bounded vertex boxes enter this integral. On these boxes the uniformly bounded sequences A_n, B_n converge weakly in L^2 to A, B. Lemma 2.10 gives strong convergence of $T_\varphi B_n$ to $T_\varphi B$, so the pairing converges to $\int A\, T_\varphi B$. This is exactly the pairing of $A(\pi_1 p)B(\pi_2 p)$ with φ.

Summing the finitely many expanded terms gives

$$\sum_{d,e} \left(\int b_d \overline{b_e} \, \mathrm{d}\nu_1 \right) \left(\int c_d \overline{c_e} \, \mathrm{d}\nu_2 \right) = \int |G|^2 \, \mathrm{d}(\nu_1 \otimes \nu_2).$$

Uniform boundedness of the products extends convergence from bounded compactly supported tests to all L^1 tests. □

Lemma 7.6 (The limiting bilinear correlation form)**.** *For almost every configuration p, there is a complex bilinear form*

$$\ell_p : L^2(\nu_0) \times L^2(\nu_1 \otimes \nu_2) \longrightarrow \mathbb{C}$$

such that

$$|\ell_p(a, G)| \le \|a\|_{L^2(\nu_0)} \|G\|_{L^2(\nu_1 \otimes \nu_2)}. \tag{7.1}$$

For trigonometric polynomials and their finite tensor sums, $\ell_p(a, G)$ is the weak- limit of $F_{0,n}[a](\pi_0 p) G_n(p)$. In particular,*

$$\ell_p(e_{s_0}, e_{s_1} \otimes e_{s_2}) = h(p; s_0, s_1, s_2).$$

Proof. Expand $F_{0,n}[a]G_n$ into three-point products and apply Lemma 7.2 to define its limit on the polynomial spaces. For every measurable configuration set B of finite measure, Cauchy–Schwarz gives

$$\left| \int_B F_{0,n}[a](\pi_0 p) G_n(p) \, \mathrm{d}\mu(p) \right| \le \left(\int_B |F_{0,n}[a](\pi_0 p)|^2 \, \mathrm{d}\mu \right)^{1/2} \left(\int_B |G_n|^2 \, \mathrm{d}\mu \right)^{1/2}.$$

Pass to the limit using Lemma 7.5. Write $l(p) = \ell_p(a, G)$, $a_0(p) = \|a\|^2_{L^2(\nu_0)}$, and $b_0(p) = \|G\|^2_{L^2(\nu_1 \otimes \nu_2)}$ for the resulting three measurable functions. For every complex number q, Cauchy–Schwarz followed by $2uv \le u^2 + v^2$ gives

$$\int_B (|q|^2 a_0 + b_0 - 2\,\mathrm{Re}(ql)) \, \mathrm{d}\mu \ge 0.$$

For each rational complex q the integrand is therefore nonnegative almost everywhere: otherwise a bounded subset of its negative set would contradict this inequality. Remove the union of the

resulting countably many null sets and then use continuity in q. The inequality holds for all q at each remaining point. If $a_0 > 0$, minimize the quadratic expression in q to obtain $|l|^2 \le a_0 b_0$; if $a_0 = 0$, varying q forces $l = 0$. This proves (7.1) without a differentiation theorem. Initially use rational shifts and rational complex coefficients, so that one null set suffices. The continuity of the limits in the shifts, and the Fourier representation in Lemma 7.3, extend the inequality to all polynomials and all finite tensor sums.

Characters with rational shifts span a dense subspace of $L^2(\nu_j)$ by Lemma 2.6. Their tensor products are dense in $L^2(\nu_1 \otimes \nu_2)$ as well. Indeed, if H is orthogonal to every $e_s \otimes e_t$ with $s, t \in \mathbb{Q}^2$, then for each s the function

$$\xi_2 \longmapsto \int H(\xi_1, \xi_2)\overline{e_s(\xi_1)}\, \mathrm{d}\nu_1(\xi_1)$$

belongs to $L^2(\nu_2)$ by Cauchy–Schwarz and is orthogonal to every rational character. It is zero almost everywhere. Countability of s and Fubini show that almost every section $H(\cdot, \xi_2)$ is orthogonal to every rational character in $L^2(\nu_1)$; hence $H = 0$ almost everywhere. Inequality (7.1) therefore gives the unique bounded bilinear extension. It also shows that its definition is independent of all polynomial or tensor representations. □

Lemma 7.7 (Infinitesimal translation and rotation invariance)**.** *Outside a fixed null set of configurations, for every $s \in \mathbb{R}^2$, $h \in \mathbb{R}$, $a \in L^2(\nu_0)$, and $G \in L^2(\nu_1 \otimes \nu_2)$,*

$$\ell_p(a, G) = \ell_p(e_s a, (e_{s+hv_\theta} \otimes e_{s+hMv_\theta})G). \tag{7.2}$$

Proof. Begin with polynomial arguments and fixed parameters s, h. Translate the configuration by

$$(x, \theta) \longmapsto (x + s/R_n, \theta + h/R_n).$$

This transformation preserves $\mathrm{d}x\, \mathrm{d}\sigma(\theta)$. For a compactly supported continuous test, the resulting change in the test tends to zero in L^1. Taylor expansion gives

$$L_j u_{\theta + h/R_n} - L_j u_\theta = h L_j v_\theta / R_n + O_M(h^2/R_n^2).$$

Multiplication by the Lipschitz bound CR_n shows that the resulting error in every input tends to zero. Thus the transformed product has the same weak-* limit as the original product, and its limiting shifts at the three vertices are respectively s, $s + hv_\theta$, and $s + hMv_\theta$ in addition to the original polynomial shifts. The variable-shift assertion of Lemma 7.2 identifies its limit and proves (7.2) for the polynomial arguments.

Take rational polynomial data and rational s, h first. Their countability gives one exceptional null set. At each remaining p, characters vary continuously in L^2 of each finite spectral measure, by dominated convergence. Inequality (7.1) therefore extends the identity to real s, h, to all polynomial data, and finally by density to both Hilbert spaces. Character multipliers have norm one throughout this extension. □

Lemma 7.8 (Atoms of one-dimensional projections)**.** *Let $z \mapsto \nu_z$ be a measurable kernel of finite positive measures on $\mathbb{R}^2$, of mass at most one, and let N be invertible. For almost every (z, θ), every positive-mass fiber of*

$$\xi \longmapsto (Nv_\theta) \cdot \xi$$

is concentrated at a single point atom of ν_z. Consequently, for almost every configuration p, the restriction of $\nu_1 \otimes \nu_2$ to

$$H_p = \{(\xi_1, \xi_2) : v_\theta \cdot \xi_1 + Mv_\theta \cdot \xi_2 = 0\} \tag{7.3}$$

is concentrated on countably many pairs of point atoms.

Proof. For distinct ξ, ξ', the equation $(Nv_\theta)\cdot(\xi-\xi')=0$ holds at only two angles modulo 2π. Lemma 2.20 makes $\mathrm{d}z\,\mathrm{d}\nu_z(\xi)\,\mathrm{d}\nu_z(\xi')$ a sigma-finite measure. Tonelli with the independent angular measure therefore gives, for almost every (z,θ),

$$\iint \mathbf{1}_{\{\xi\neq\xi',\,(Nv_\theta)\cdot(\xi-\xi')=0\}}\,\mathrm{d}\nu_z(\xi)\,\mathrm{d}\nu_z(\xi')=0.$$

For completeness, integrate first over bounded sets of z and then exhaust $\mathbb{R}^2$. On a positive-mass fiber the product measure is thus supported on the diagonal. A finite positive measure with this property is a point mass: otherwise some Borel subset has measure strictly between zero and the total mass, and its product with its complement gives positive off-diagonal mass.

Apply this conclusion to $N=I$ at vertex 1 and to $N=M$ at vertex 2. The map $(x,\theta)\mapsto(\pi_j p,\theta)$ preserves measure, so the conclusions hold for almost every p simultaneously. Let η_1 and η_2 be the pushforwards of ν_1 and ν_2 by $v_\theta\cdot\xi$ and $Mv_\theta\cdot\xi$. The product mass of H_p is

$$\sum_{t:\,\eta_1(\{t\})>0}\eta_1(\{t\})\eta_2(\{-t\}).$$

Only countably many terms can be positive: a finite measure has finitely many atoms of mass at least $1/n$ for each $n\geq 1$. By the first part, each positive term is carried by one pair of point atoms. This proves the assertion about the restricted measure, not just its total mass. □

Lemma 7.9 (Countably many measurable selectors cover all atoms)**.** *For a measurable kernel $z\mapsto\nu_z$ of finite positive Borel measures on $\mathbb{R}^2$, there is a countable family of measurable maps $a_m:\mathbb{R}^2\to\mathbb{R}^2$ such that every point atom of every nonzero ν_z occurs among the values $a_m(z)$.*

Proof. Where the mass is nonzero, normalize ν_z to a probability measure. For each of its two coordinate marginals and every $r\in\mathbb{Q}\cap(0,1)$, form the quantile

$$q_{k,r}(z)=\inf\{t\in\mathbb{R}:\nu_z(\{\xi:\xi_k\leq t\})/\nu_z(\mathbb{R}^2)\geq r\},\qquad k=1,2.$$

These values are finite, since the normalized distribution functions tend to zero and one at the two ends of the real line. They are measurable: on the set $m(z)=\nu_z(\mathbb{R}^2)>0$,

$$\{z:q_{k,r}(z)<a\}=\bigcup_{\substack{t\in\mathbb{Q}\\ t<a}}\{z:\nu_z(\{\xi:\xi_k\leq t\})\geq r\,m(z)\}.$$

Indeed, the threshold set defining the infimum is an upper interval, so it meets $(-\infty,a)$ precisely when it contains a rational point less than a. Every set in the displayed union is measurable. Take every pair $(q_{1,r},q_{2,s})$ with rational $r,s\in(0,1)$, and set these maps to zero where $\nu_z=0$.

If (a,b) is a point atom, then a and b are atoms of the respective marginals. At either coordinate the distribution function has a positive jump. Every quantile level strictly inside that jump returns the coordinate in question. Choose a rational level inside each jump. The resulting pair is (a,b). It is harmless that other pairs of quantiles need not be atoms. □

Lemma 7.10 (Rigidity for all contributing atomic triples)**.** *For almost every configuration p, every triple of point atoms ξ_j of $\nu_{j,\pi_j p}$ satisfying*

$$\xi_0+\xi_1+\xi_2=0,\qquad v_\theta\cdot(\xi_1+M^T\xi_2)=0 \tag{7.4}$$

also satisfies $\xi_1+M^T\xi_2=0$.

Proof. Apply Lemma 7.9 to the three spectral kernels. There are countably many triples of the resulting measurable vector fields. Fix one such triple a_0,a_1,a_2 and let E be the measurable set where its values at the vertices satisfy (7.4) but $a_1(\pi_1 p)+M^T a_2(\pi_2 p)\neq 0$. If $\mu(E)>0$, restrict to a positive-measure subset on which x and all three field values are bounded. This is possible by a countable exhaustion. If the respective bounds are L and K, take $X_j=\{z:|z|\leq L+\|L_j\|,\ |a_j(z)|\leq K\}$. All vertices of this subset then lie in the bounded measurable sets X_j, and the fields are bounded on

them. Lemma 5.3 provides a positive-measure core in this subset. The restrictions of the fields to its vertex sets are square integrable, and they satisfy the stationary equations there. Theorem 6.5 forces $a_1(\pi_1 p) + M^T a_2(\pi_2 p) = 0$ almost everywhere on that core, a contradiction. Hence $\mu(E) = 0$.

Remove the union of these exceptional sets over the countable family of triples. At every remaining configuration, each triple of point atoms is represented by a triple of the selectors, so the asserted implication holds for all atomic triples at once. □

Lemma 7.11 (Invariance under displacement smoothing)**.** *For almost every configuration p and every $y \in \mathbb{R}^2$,*

$$\ell_p(1, e_y \otimes e_{My}) = \ell_p(1, 1).$$

Proof. Fix p outside all the exceptional sets in the preceding lemmas. In (7.2), put $s = 0$ and average h over $[0, H]$. The resulting multiplier on the second Hilbert space is

$$\frac{1}{H} \int_0^H \mathrm{e}\big(h(v_\theta \cdot \xi_1 + M v_\theta \cdot \xi_2)\big)\, \mathrm{d}h.$$

It is bounded by one and converges pointwise to $\mathbf{1}_{H_p}$. Dominated convergence gives strong convergence after multiplication by any $G \in L^2(\nu_1 \otimes \nu_2)$. Boundedness of ℓ_p therefore gives

$$\ell_p(a, G) = \ell_p(a, \mathbf{1}_{H_p} G). \tag{7.5}$$

By Lemma 7.8, the measure on H_p is carried by countably many pairs of point atoms. For one such pair (ξ_1, ξ_2), let E_{ξ_1,ξ_2} be the indicator of that point in the product frequency space. Put $h = 0$ in (7.2) and average s over $[0, H]^2$. The scalar character contributed by this second argument can be moved into the first argument by bilinearity. Strong dominated convergence in $L^2(\nu_0)$ then gives

$$\ell_p(a, E_{\xi_1,\xi_2}) = \ell_p(a \mathbf{1}_{\{-\xi_1-\xi_2\}}, E_{\xi_1,\xi_2}).$$

This is zero unless $\xi_0 = -\xi_1 - \xi_2$ is a point atom of ν_0. In the remaining case the three atoms satisfy (7.4), so Lemma 7.10 gives $\xi_1 + M^T \xi_2 = 0$. Consequently

$$e_y(\xi_1) e_{My}(\xi_2) = \mathrm{e}\big(y \cdot (\xi_1 + M^T \xi_2)\big) = 1.$$

Approximate $\mathbf{1}_{H_p} G$ by its restrictions to finitely many of the atomic pairs. The approximation converges in the product L^2 space, and (7.1) passes the preceding identity to the limit. Thus, in fact,

$$\ell_p(a, (e_y \otimes e_{My}) G) = \ell_p(a, G)$$

for every a, G, y. Taking $a = 1$ and $G = 1$ proves the assertion. □

Proposition 7.12 (Cancellation for bounded Lipschitz inputs)**.** *Let $\Psi \in L^1(\mathbb{R}^2)$ satisfy $\int \Psi = 0$, and put $\Psi_R(u) = R^2 \Psi(Ru)$. For every bounded box Q and every $C < \infty$,*

$$\sup_{\substack{|f_j| \le \mathbf{1}_Q \\ \mathrm{Lip}(f_j) \le CR}} |\Lambda_{\sigma * \Psi_R}(f_0, f_1, f_2)| \longrightarrow 0 \qquad (R \to \infty).$$

Proof. It suffices to consider an arbitrary sequence of admissible inputs with $R_n \to \infty$ and find a subsequence on which the forms tend to zero. Otherwise a sequence on which their absolute values are bounded below would contradict that property. Take the subsequence and correlation forms constructed above. Choose $w \in C_c(\mathbb{R}^2)$ with $w = 1$ on Q. The convolution identity in Lemma 4.5 gives

$$\begin{aligned} &\Lambda_{\sigma * \Psi_{R_n}}(f_{0,n}, f_{1,n}, f_{2,n}) \\ &\quad = \int \Psi(y) \int w(x) f_{0,n}(x) f_{1,n}(x + u_\theta + y/R_n) f_{2,n}(x + M u_\theta + My/R_n)\, \mathrm{d}\mu(p)\, \mathrm{d}y. \end{aligned}$$

For each fixed y the inner integral tends, by Lemma 7.6, to

$$\int w(x) \ell_p(1, e_y \otimes e_{My})\, \mathrm{d}\mu(p).$$

All these integrals are bounded by $\|w\|_1$, independently of n, y. Dominated convergence with $\Psi \in L^1$ is therefore applicable. Lemma 7.11 makes the limit independent of y, and $\int \Psi = 0$ makes the resulting expression zero. □

8. Localization and a global annular estimate

Lemma 8.1 (From bounded local inputs to endpoint operator norms)**.** *Let $\mu_n = \sigma * \kappa_n$, where $\sup_n \|\kappa_n\|_{\mathrm{TV}} < \infty$, and suppose that for some fixed $U < \infty$,*

$$|\mu_n|(\{u : |u| > U\}) \longrightarrow 0.$$

Let P_n be convolution by measures supported in the closed unit ball, with uniformly bounded total variation. Suppose that for every fixed triple $\chi_0, \chi_1, \chi_2 \in C_c^\infty(\mathbb{R}^2)$,

$$\sup_{\|f_j\|_\infty \le 1} |\Lambda_{\mu_n}(\chi_0 P_n f_0, \chi_1 P_n f_1, \chi_2 P_n f_2)| \longrightarrow 0. \tag{8.1}$$

Then the form $\Lambda_{\mu_n}(P_n f_0, P_n f_1, P_n f_2)$ has operator norm tending to zero on every permutation of $L^2 \times L^2 \times L^\infty$.

Proof. Choose a nonnegative smooth partition of unity $\sum_{q \in \mathbb{Z}^2} \chi_{0,q}(x) = 1$, where the functions are translates of one compactly supported function. For $j = 1, 2$ choose fixed compactly supported smooth cutoffs, between zero and one, whose translates $\chi_{j,q}$ equal one at $x + L_j u$ whenever $x \in \operatorname{supp} \chi_{0,q}$ and $|u| \le U$. Thus

$$0 \le \sum_q \prod_{j=0}^{2} \chi_{j,q}(x + L_j u) \le 1,$$

and this sum is one when $|u| \le U$. The total-variation endpoint estimate of Lemma 4.4 bounds the error in inserting this sum by

$$o_n(1) \|f_i\|_\infty \prod_{j \ne i} \|f_j\|_2$$

for any fixed placement i of the bounded input. The uniform convolution bounds for P_n have been absorbed into the error.

For each localized term, restrict the two L^2 inputs to boxes $B_j + q$ containing the closed unit neighborhoods of the supports of $\chi_{j,q}$. This restriction is exact because the convolution measure defining P_n is supported in the unit ball. Denote their local L^2 norms by $r_{j,q}$. If either is zero, the term is zero. Otherwise divide the two restricted inputs by these norms, and divide the bounded input by its global supremum norm. It is enough to bound the resulting localized expression when the two local L^2 norms and the global L^∞ norm are one.

Fix $T \ge 1$. For either normalized L^2 input g, write

$$g = g\mathbf{1}_{\{|g| \le T\}} + g\mathbf{1}_{\{|g| > T\}}.$$

The second term has L^2 norm at most one and is supported on a set of area at most T^{-2}. In a form containing this term, expand its convolution under P_n as an integral of translates. Each translated term, after multiplication by the cutoff, still has support of area at most T^{-2} and L^2 norm at most one. Lemma 4.11, including its assertion for $\sigma * \kappa_n$, gives an error

$$C\omega_M(T^{-2})$$

uniformly in n and q. The other projected L^2 input and the projected bounded input have uniformly bounded norms. The same argument removes the large part of the other L^2 input, with the same type of bound.

The two remaining inputs have absolute value at most T. Hypothesis (8.1) now bounds the normalized localized form by

$$C\omega_M(T^{-2}) + T^2 \varepsilon_n, \qquad \varepsilon_n \longrightarrow 0.$$

The number ε_n is independent of q: simultaneous translation of the inputs and cutoffs reduces each q to the same fixed cutoff triple, since P_n and Λ_{μ_n} commute with simultaneous translations.

Restore the local norms. If j, k are the two L^2 indices, Cauchy–Schwarz and bounded overlap of the boxes give

$$\sum_q r_{j,q} r_{k,q} \le \left(\sum_q r_{j,q}^2\right)^{1/2} \left(\sum_q r_{k,q}^2\right)^{1/2} \le C\|f_j\|_2 \|f_k\|_2.$$

This also justifies the sum of localized forms. First let $n \to \infty$ with T fixed, and then let $T \to \infty$. The small-support modulus tends to zero, which proves the claimed endpoint norm convergence for this placement. The argument applies to each of the three choices of i. □

Corollary 8.2 (Global cancellation on one annulus)**.** *Let $\psi \in C_c^\infty(\mathbb{R}^2)$ be supported in a compact annulus disjoint from the origin. Put*

$$K_R = \sigma * (R^2 \check{\psi}(R\,\cdot)).$$

For every fixed $0 < C_0 < \infty$ there is a function $a(R) \to 0$, with $0 \le a(R) \le \|\check{\psi}\|_1$, such that whenever $f_i \in L^\infty$ and $f_j \in L^2$ for $j \ne i$, and all three inputs have Fourier supports in $\overline{B(0, C_0 R)}$,

$$|\Lambda_{K_R}(f_0, f_1, f_2)| \le a(R)\|f_i\|_\infty \prod_{j \ne i} \|f_j\|_2, \qquad i = 0, 1, 2.$$

For the bounded input, Fourier support is understood through the test-function condition in Definition 2.7.

Proof. Choose a Schwartz function p whose Fourier transform is one on a neighborhood of $\overline{B(0, C_0)}$. Then $p_R = R^2 p(R\,\cdot)$ reproduces each of the specified inputs by Lemma 2.8, applied at scale R.

Choose $\chi \in C_c^\infty(B(0,1))$ equal to one near zero and put $q_R = \chi p_R$. Rapid decay gives

$$\|q_R - p_R\|_1 = o(1), \qquad \sup_R \|q_R\|_1 < \infty, \qquad \|\nabla q_R\|_1 = O(R).$$

Apply Lemma 8.1 along an arbitrary sequence $R_n \to \infty$ with $P_n f = q_{R_n} * f$ and $\mu_n = K_{R_n}$. Its convolution factor $\kappa_n = R_n^2 \check{\psi}(R_n\,\cdot)$ has bounded L^1 norm. Moreover,

$$|K_{R_n}|(\{|u| > 2\}) \le \int_{|y|>1} R_n^2 |\check{\psi}(R_n y)|\, \mathrm{d}y \longrightarrow 0.$$

For fixed smooth compactly supported cutoffs and bounded inputs, the functions $\chi_j P_n f_j$ have common bounded support, uniformly bounded magnitude, and Lipschitz constants $O(R_n)$. The integral of $\check{\psi}$ is $\psi(0) = 0$, so Proposition 7.12 proves the hypothesis (8.1). The localized lemma consequently gives endpoint norm convergence with q_R in all input slots.

Finally replace q_R by p_R in the three slots. Young's inequality gives an operator difference at most $\|q_R - p_R\|_1 = o(1)$ on both L^2 and L^∞. The uniform total-variation bound on K_R and Lemma 4.4 show that the replacement costs $o(1)$ in each endpoint form norm. Since p_R reproduces the inputs, this proves the stated estimate. Take $a(R)$ to be the maximum of the actual restricted endpoint norms over the three placements. The total-variation bound gives $a(R) \le \|\check{\psi}\|_1$, and the argument proves $a(R) \to 0$. □

9. Summation over frequency annuli

Throughout this section, the annular estimate of Corollary 8.2 is used for each of the three choices of the bounded input. All constants below may depend on M and on the smooth functions fixed in this section.

Definition 9.1 (Annular kernels)**.** Fix a real radial function $\chi \in C_c^\infty(\mathbb{R}^2)$ which equals one on $B(0,1)$ and vanishes outside $B(0,2)$. Define

$$\psi(b) = \chi(b/2) - \chi(b), \qquad q_R(u) = R^2 \check{\chi}(Ru), \qquad k_r(u) = r^2 \check{\psi}(ru).$$

Thus ψ is supported in $\{1 \le |b| \le 4\}$ and $\int q_R = \chi(0) = 1$. Write

$$\Lambda_r = \Lambda_{\sigma * k_r}.$$

The identities $k_r = q_{2r} - q_r$ and $\|k_r\|_1 = \|\check{\psi}\|_1$ hold for every $r > 0$.

Lemma 9.2 (Frequency representation with compact spectral supports)**.** *Suppose that $f_0, f_1, f_2 \in L^2(\mathbb{R}^2)$ have compact Fourier supports. Then*

$$\Lambda_r(f_0, f_1, f_2) = \iint \widehat{f_0}(-\xi_1 - \xi_2)\widehat{f_1}(\xi_1)\widehat{f_2}(\xi_2) \cdot \widehat{\sigma}(\xi_1 + M^T\xi_2)\psi((\xi_1 + M^T\xi_2)/r)\, \mathrm{d}\xi_1\, \mathrm{d}\xi_2.$$

The double integral is absolutely convergent.

Proof. For Schwartz functions, this is Lemma 4.7, applied with the kernel $\sigma * k_r$ of Definition 9.1.

For fixed ξ_2, Cauchy–Schwarz gives

$$\int |\widehat{f_0}(-\xi_1 - \xi_2)\widehat{f_1}(\xi_1)|\, \mathrm{d}\xi_1 \le \|f_0\|_2 \|f_1\|_2.$$

The compact support and the L^2 integrability of $\widehat{f_2}$ imply $\widehat{f_2} \in L^1$. Consequently the absolute value of the double integral is bounded by

$$\|\widehat{\sigma}\psi(\cdot/r)\|_\infty \|f_0\|_2 \|f_1\|_2 \|\widehat{f_2}\|_1.$$

Approximate the three Fourier transforms by smooth compactly supported functions, with all approximants supported in fixed compact enlargements of their original supports. They converge in L^2 and, for the third input, also in L^1. The displayed bound passes the Fourier integral to the limit. Lemma 4.10, applied after convolution with k_r, passes the physical-space form to the same limit. □

Definition 9.3 (Common frequency coordinates)**.** Put $A = M^T$ and define invertible linear maps

$$T_0 = (A - I)^{-1}, \qquad T_1 = -A^{-1}, \qquad T_2 = I.$$

For a frequency triple with $\xi_0 + \xi_1 + \xi_2 = 0$, write $z_j = T_j\xi_j$ and $b = \xi_1 + A\xi_2$.

Lemma 9.4 (Frequency separation and modulation invariance)**.** *For the coordinates of Definition 9.3,*

$$z_0 - z_2 = -(A - I)^{-1}b, \quad z_1 - z_2 = -A^{-1}b, \quad z_1 - z_0 = A^{-1}(A - I)^{-1}b.$$

In particular, whenever $\psi(b/r) \neq 0$, every pairwise distance $|z_i - z_j|$ is bounded above and below by positive multiples of r. For $w \in \mathbb{R}^2$ set

$$v_0 = (A - I)w, \qquad v_1 = -Aw, \qquad v_2 = w.$$

Replacing each $f_j(x)$ by $\mathrm{e}(-v_j \cdot x)f_j(x)$ leaves $\Lambda_\nu(f_0, f_1, f_2)$ unchanged for every finite measure ν. This replacement translates all three z-frequency supports by $-w$.

Proof. Substitute $\xi_0 = -\xi_1 - \xi_2$ in the definition of z_0 and subtract. The matrices A and $A - I$, and hence their inverses, commute. This gives the three displayed identities. Each matrix on the right is invertible, so each norm is comparable to $|b|$. The support condition in Definition 9.1 gives $r \le |b| \le 4r$.

The modulation factors in the integrand of the form multiply to

$$\mathrm{e}(-x \cdot (v_0 + v_1 + v_2) - u \cdot (v_1 + M^T v_2)) = 1.$$

Indeed both vectors in parentheses vanish. Finally $T_j v_j = w$ for every j, so the asserted translation of supports follows from the Fourier transform of a modulation. □

Definition 9.5 (Smooth frequency partitions and spectral mass)**.** Choose a nonnegative $\varphi \in C_c^\infty(\mathbb{R}^2)$ with $\sum_{q\in\mathbb{Z}^2} \varphi(z-q) = 1$. Fix a closed square Q_* containing its support. For a small fixed $h > 0$, define

$$\widehat{P_{j,q,r} f}(\xi) = \varphi(T_j\xi/(hr) - q)\,\widehat{f}(\xi).$$

Let $Q_{q,r} = hr(q + Q_*)$. For a fixed triple of L^2 inputs define the finite positive measure

$$\mu = \sum_{j=0}^{2} (T_j)_*(|\widehat{f_j}(\xi)|^2\,\mathrm{d}\xi).$$

A triple (q_0, q_1, q_2) is called contributing at scale r if there are frequencies ξ_j with $\sum_j \xi_j = 0$, with $T_j\xi_j \in Q_{q_j,r}$ for all j, and with $\psi((\xi_1 + A\xi_2)/r) \neq 0$.

Lemma 9.6 (Properties of the smooth partition)**.** *The parameter h can be chosen depending only on M and φ so that the following statements hold uniformly in $r > 0$.*

(1) The square family $\{Q_{q,r}\}$ has bounded overlap. The three squares in every contributing triple have mutual distance at least dr and their union has diameter at most Dr, for fixed $d, D > 0$. Every square in a specified slot occurs in at most K contributing triples.

(2) There is a constant C such that

$$\|P_{j,q,r} f_j\|_2^2 \le C\mu(Q_{q,r}), \qquad \|P_{j,q,r} f_j\|_\infty \le C\|f_j\|_\infty, \qquad \sum_q \|P_{j,q,r} f_j\|_2^2 \le C\|f_j\|_2^2.$$

(3) For L^2 inputs the following expansion is absolutely convergent:

$$\Lambda_r(f_0, f_1, f_2) = \sum_{(q_0,q_1,q_2)\,\text{contributing}} \Lambda_r(P_{0,q_0,r} f_0, P_{1,q_1,r} f_1, P_{2,q_2,r} f_2).$$

Proof. The square side length is a fixed multiple of hr, and its centers form a lattice of mesh hr. This proves bounded overlap. By Lemma 9.4, the witnessing points of a contributing triple have pairwise distances between $c_M r$ and $C_M r$. Choose h small enough that the diameters of the support squares are less than $c_M r/4$. Every pair of entire squares is then separated by a positive multiple of r, and the union lies in a ball of radius a fixed multiple of r. Once one square is fixed, the centers of either of its partners lie within $O_M(r)$. There are only $O_{M,h}(1)$ lattice centers in this region, which bounds the number of partner pairs.

Plancherel gives the first L^2 bound, since the multiplier vanishes outside $T_j^{-1}Q_{q,r}$ and has uniformly bounded magnitude. The same argument followed by the bounded-overlap estimate proves the sum of squares bound. A translation, dilation, and fixed invertible linear change of variables in frequency show that the convolution kernel of $P_{j,q,r}$ has L^1 norm independent of q and r. Young's inequality gives the L^∞ estimate.

Put $u_{j,q} = \|P_{j,q,r} f_j\|_2$. By the convolved version of Lemma 4.10, each summand has absolute value at most $C u_{0,q_0} u_{1,q_1} u_{2,q_2}$. The constant is independent of r because $\|k_r\|_1$ is fixed. The bounded partner count and Cauchy–Schwarz yield

$$\sum_{\text{contributing}} u_{0,q_0} u_{1,q_1} u_{2,q_2} \le \Big(\sup_q u_{2,q}\Big)\Big(\sum_{\text{contributing}} u_{0,q_0}^2\Big)^{1/2}\Big(\sum_{\text{contributing}} u_{1,q_1}^2\Big)^{1/2}$$
$$\le C\prod_{j=0}^{2} \|f_j\|_2.$$

This proves absolute convergence.

Finite partial partition sums converge to each input in L^2, by Plancherel and dominated convergence. For finite sums, expand the form by trilinearity. Lemma 9.2 applies to every summand,

whose Fourier support is compact, and shows that every noncontributing term vanishes. Pass to the limit using Lemma 4.10 on the left and the just-proved absolute convergence on the right. □

Lemma 9.7 (One frequency triple)**.** *Suppose that $\|f_j\|_2, \|f_j\|_\infty \le B$. For a contributing triple let $m \le n \le p$ be the three numbers $\mu(Q_{q_j,r})$, in increasing order. For a sufficiently large fixed frequency-ball constant in Corollary 8.2, its modulus $a(r)$ satisfies*

$$|\Lambda_r(P_{0,q_0,r}f_0, P_{1,q_1,r}f_1, P_{2,q_2,r}f_2)| \le C_B a(r)^{1/2} np^{1/4}.$$

Proof. Lemma 4.10 and Lemma 9.6 first give $C(mnp)^{1/2}$. Translate all three z-frequency squares by the negative of the center of one square. Lemma 9.4 realizes this as a form-preserving modulation. Lemma 9.6 shows that the translated supports lie in a fixed ball of radius $C_M r$ in each original frequency plane. Apply Corollary 8.2, putting the input corresponding to mass p in the bounded slot. Its supremum norm is at most CB, and the other two L^2 norms are at most $C\sqrt{m}$ and $C\sqrt{n}$. This gives the second bound $C_B a(r)(mn)^{1/2}$.

The minimum of two nonnegative bounds is at most their geometric mean. The resulting estimate is $C_B a(r)^{1/2}(mn)^{1/2}p^{1/4}$, which is at most the claimed bound because $m \le n$. If a mass is zero, the relevant projected input is zero in L^2 and the conclusion holds directly. □

Definition 9.8 (An increasing mass functional)**.** For a finite positive Borel measure μ on $\mathbb{R}^2$ and $t > 0$, put

$$H(t) = \int_{\mathbb{R}^2} \mu(B(z,t))^{1/4}\, \mathrm{d}\mu(z).$$

This is nondecreasing and satisfies $0 \le H(t) \le \mu(\mathbb{R}^2)^{5/4}$.

Lemma 9.9 (Geometric mass increment)**.** *For the square families and contributing triples of Lemma 9.6, there are fixed constants $0 < c < C$ such that*

$$\sum_{\text{contributing}} np^{1/4} \le C_M\big(H(Cr) - H(cr)\big),$$

where $m \le n \le p$ are the masses of the three squares in each triple.

Proof. Choose $c < d/4$, with d from Lemma 9.6. For a fixed triple, the neighborhoods $N_j = Q_{q_j,r} + B(0,cr)$ are pairwise disjoint. Choose a ball B_* containing their union. Since the diameter of that union is $O_M(r)$, the constant C can be fixed so that $B_* \subset B(z, Cr)$ for every z in any of the three squares. At least two of the disjoint neighborhoods have mass at most $\mu(B_*)/2$. Select their indices i, j.

For $z \in Q_{q_i,r} \cup Q_{q_j,r}$, the smaller ball $B(z,cr)$ is contained in the corresponding selected neighborhood. Therefore

$$\begin{aligned}\mu(B(z,Cr))^{1/4} - \mu(B(z,cr))^{1/4} &\ge (1 - 2^{-1/4})\mu(B_*)^{1/4} \\ &\ge (1 - 2^{-1/4})p^{1/4}.\end{aligned}$$

The selected squares are disjoint, and their combined mass is at least n, irrespective of which two indices were selected. Integrating the last inequality over their union bounds $np^{1/4}$ by a fixed multiple of that integral.

Sum over contributing triples. Each point belongs to only boundedly many support squares, and each square has only boundedly many partner pairs. Hence the multiplicity of the selected regions is uniformly bounded. The integrand is nonnegative, so the sum is at most a constant times its integral over the whole plane, which is $H(Cr) - H(cr)$. □

Lemma 9.10 (Dyadic telescoping of the mass increments)**.** *For $0 < c < C$, every nondecreasing bounded function H satisfies*

$$\sum_{k=0}^{\infty} \big(H(C2^k R) - H(c2^k R)\big) \le D \sup_{t>0} H(t),$$

provided $H \geq 0$ and D is any positive integer with $2^D c \geq C$.

Proof. By monotonicity, the summand is at most $H(c2^{k+D}R) - H(c2^k R)$. In a finite sum over $0 \leq k \leq N$, all but the first and last D terms cancel. The terms with a minus sign are nonpositive and may be discarded, leaving at most $D \sup H$. The original summands are nonnegative, so monotone convergence gives the infinite-sum bound. □

Proposition 9.11 (Uniform annular tails)**.** *Let $R \geq 1$ be dyadic. If $\|f_j\|_2, \|f_j\|_\infty \leq B$, then*

$$\sum_{r=2^k \geq R} |\Lambda_r(f_0, f_1, f_2)| \leq C_{M,B} \left(\sup_{r \geq R} a(r) \right)^{1/2}.$$

In particular this tail tends to zero as $R \to \infty$, uniformly over these inputs.

Proof. Combine the absolutely convergent expansion in Lemma 9.6, the estimate in Lemma 9.7, and Lemma 9.9. At each scale the result is

$$|\Lambda_r(f_0, f_1, f_2)| \leq C_{M,B} a(r)^{1/2} (H(Cr) - H(cr)).$$

The same spectral measure μ is used at every scale, and Plancherel shows that $\mu(\mathbb{R}^2) = \sum_j \|f_j\|_2^2 \leq 3B^2$. Pull out the supremum of $a(r)^{1/2}$ and apply Lemma 9.10 and Definition 9.8. Finally, $a(r) \to 0$ implies $\sup_{r \geq R} a(r) \to 0$. □

10. Approximation of the counting form

Lemma 10.1 (Exhaustion by smooth low frequencies)**.** *For every finite measure κ and every triple of L^2 inputs,*

$$\Lambda_{\sigma * \kappa * q_S}(f_0, f_1, f_2) \longrightarrow \Lambda_{\sigma * \kappa}(f_0, f_1, f_2) \qquad (S \to \infty).$$

Proof. Write $q_S(u) = S^2 \check{\chi}(Su)$ and use Lemma 4.5. Convolution by q_S averages simultaneous translations of the inputs at vertices 1 and 2 by y/S and My/S, with weight $\check{\chi}(y)\,dy$. For each fixed y, these inputs converge to the originals in L^2. The convolved version of Lemma 4.10 therefore gives convergence of the corresponding form. The absolute value is bounded uniformly in y and S by

$$C_M \|\kappa\|_{\mathrm{TV}} \prod_j \|f_j\|_2.$$

Dominated convergence against $|\check{\chi}(y)|\,dy$ applies, and $\int \check{\chi} = 1$ gives the stated limit. No positivity of $\check{\chi}$ is required. □

Lemma 10.2 (Tails after convolution)**.** *For every finite measure κ, dyadic $R \geq 1$, and inputs with $\|f_j\|_2, \|f_j\|_\infty \leq B$,*

$$\sum_{r=2^k \geq R} |\Lambda_{\sigma * \kappa * k_r}(f_0, f_1, f_2)| \leq C_{M,B} \|\kappa\|_{\mathrm{TV}} \left(\sup_{r \geq R} a(r) \right)^{1/2}.$$

Moreover the sum without absolute values equals $\Lambda_{\sigma * \kappa}(f_0, f_1, f_2) - \Lambda_{\sigma * \kappa * q_R}(f_0, f_1, f_2)$.

Proof. Lemma 4.5 expresses each summand as an integral against κ of Λ_r applied to f_0 and translates of f_1, f_2. These translations preserve the two norms. Apply Proposition 9.11, first to finite sums, integrate against $|\kappa|$, and then use monotone convergence.

For a finite consecutive dyadic sum the kernel identity in Definition 9.1 gives

$$\sum_{k=0}^{N} \sigma * \kappa * k_{2^k R} = \sigma * \kappa * (q_{2^{N+1} R} - q_R).$$

Apply Lemma 10.1 and the absolute convergence just proved to pass to the limit. □

Proposition 10.3 (Smoothing for inputs bounded in both norms)**.** *Let $\rho \in L^1(\mathbb{R}^2)$ be nonnegative with $\int \rho = 1$, and set $\rho_\varepsilon(u) = \varepsilon^{-2}\rho(u/\varepsilon)$. For each fixed $B < \infty$,*

$$\sup_{\|f_j\|_2, \|f_j\|_\infty \le B} |\Lambda_{\sigma * \rho_\varepsilon - \sigma}(f_0, f_1, f_2)| \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Proof. Fix a dyadic $R \ge 1$. Apply Lemma 10.2 to $\kappa = \rho_\varepsilon(u)\,\mathrm{d}u$ and to $\kappa = \delta_0$. Each has total variation one. Thus the two high-frequency tails contribute together at most

$$2C_{M,B}\left(\sup_{r \ge R} a(r)\right)^{1/2}.$$

Their remaining difference has kernel $\sigma * (\rho_\varepsilon * q_R - q_R)$. For fixed R, translation continuity in L^1 and dominated convergence give

$$\|\rho_\varepsilon * q_R - q_R\|_1 \le \int \rho(y) \|q_R(\,\cdot - \varepsilon y) - q_R\|_1 \,\mathrm{d}y \longrightarrow 0.$$

The integrand is bounded by $2\|q_R\|_1 \rho(y)$, so this argument requires only the stated L^1 assumptions on ρ. The total variation estimate in Lemma 4.4 bounds the low-frequency form by $B^3\|\rho_\varepsilon * q_R - q_R\|_1$.

Given a desired error, first choose R sufficiently large to make the tail bound small, and then choose ε sufficiently small to make this last term small. The estimates are uniform in the inputs. □

Theorem 10.4 (Uniform smoothing in the endpoint form norm)**.** *For every probability density $\rho \in L^1(\mathbb{R}^2)$,*

$$\lim_{\varepsilon \downarrow 0} \sup_{s > 0} \|\Lambda_{(\sigma * \rho_\varepsilon - \sigma)_s}\| = 0.$$

The same conclusion holds for every permutation of the $L^2 \times L^2 \times L^\infty$ norm.

Proof. Take an arbitrary sequence $\varepsilon_n \downarrow 0$ and put

$$\kappa_n = \rho_{\varepsilon_n}(u)\,\mathrm{d}u - \delta_0, \qquad \mu_n = \sigma * \kappa_n.$$

Then $\|\kappa_n\|_{\mathrm{TV}} \le 2$, and

$$|\mu_n|(\{|u| > 2\}) \le \int_{|w|>1} \rho_{\varepsilon_n}(w)\,\mathrm{d}w = \int_{|y| > 1/\varepsilon_n} \rho(y)\,\mathrm{d}y \longrightarrow 0.$$

Use Lemma 8.1 with P_n equal to the identity. For every fixed triple of smooth compactly supported cutoffs χ_j and every $\|f_j\|_\infty \le 1$, the functions $\chi_j f_j$ have L^2 norms at most $\|\chi_j\|_2$ and supremum norms at most $\|\chi_j\|_\infty$. Proposition 10.3 therefore verifies its localized hypothesis, with a bound B depending only on these fixed cutoffs. The lemma gives convergence of the endpoint form norm to zero, for each of the three bounded slots. Since the sequence ε_n was arbitrary, convergence holds as $\varepsilon \downarrow 0$.

Finally, dilation invariance from Lemma 4.4 gives

$$\|\Lambda_{(\sigma * \rho_\varepsilon - \sigma)_s}\| = \|\Lambda_{\sigma * \rho_\varepsilon - \sigma}\| \qquad (s > 0).$$

This proves the asserted uniformity in s. □

11. From smoothing to triangles

Lemma 11.1 (Continuity under small changes of scale)**.** *For every endpoint placement i,*

$$\lim_{r \to 1} \|\Lambda_{\sigma_r - \sigma}\|_i = 0.$$

Proof. Fix a nonnegative smooth compactly supported probability density ρ. For $t > 0$ let $k_t = \sigma * \rho_t$, where $\rho_t(u) = t^{-2}\rho(u/t)$. This is an L^1 density. The identity

$$\sigma_r - \sigma = (\sigma - k_t)_r + ((k_t)_r - k_t) + (k_t - \sigma)$$

and Lemma 4.4 give

$$\|\Lambda_{\sigma_r-\sigma}\|_i \le 2\|\Lambda_{\sigma-k_t}\|_i + \|(k_t)_r - k_t\|_1.$$

Here $(k_t)_r(u) = r^{-2}k_t(u/r)$ as a density. With t fixed, the last norm tends to zero by continuity of dilations in L^1. Theorem 10.4 makes the first term arbitrarily small as $t \downarrow 0$. Taking these limits in that order proves the assertion. □

Definition 11.2 (Density and triangle counts)**.** For a Lebesgue measurable set $E \subset \mathbb{R}^2$, its upper Banach density is

$$d^*(E) = \limsup_{L\to\infty} \sup_{z\in\mathbb{R}^2} \frac{|E \cap (z + [0,L]^2)|}{L^2}.$$

For a bounded, compactly supported, nonnegative function f and $s > 0$, write

$$D_s(f) = \Lambda_{\sigma_s}(f,f,f) = \int_{\mathbb{T}} \int_{\mathbb{R}^2} f(x)f(x+su_\theta)f(x+sMu_\theta)\,\mathrm{d}x\,\mathrm{d}\sigma(\theta).$$

A scale s is *missed by* E if there are no $x \in \mathbb{R}^2$, $\theta \in \mathbb{T}$ for which all three displayed vertices belong to E.

Lemma 11.3 (Large dense squares and exact copies)**.** *If $0 < \delta < d^*(E)$, there are squares Q of arbitrarily large side length for which $|E \cap Q| \ge \delta|Q|$. For each such square there is a Borel set $A \subset E \cap Q$ of the same measure. If s is missed by E, then $D_s(\mathbf{1}_A) = 0$. If $D_s(\mathbf{1}_A) > 0$, then A, and hence E, contains a translate of a rotation of $s\{0, e_1, Me_1\}$.*

Proof. Choose δ' strictly between δ and $d^*(E)$. The definition of limsup supplies arbitrarily large L for which the supremum over z of the density in $z + [0,L]^2$ exceeds δ'. Approximate this supremum closely enough to obtain density at least δ. Existence of the Borel subset follows from completeness and regularity of Lebesgue measure.

The integrand defining $D_s(\mathbf{1}_A)$ is a nonnegative Borel function. It is identically zero when the scale is missed. If its integral is positive, it equals one at some configuration. The vertices there have the form $x + s\{0, u_\theta, Mu_\theta\}$, which is the required rotated triangle because M commutes with rotations. □

Definition 11.4 (Word notation for the geometric application)**.** For $n \ge 1$, use $\mathcal{W}_n = \{0,1,2\}^n$, obtained from $[3]^n$ by the relabeling $j \mapsto j-1$. Words, combinatorial lines, their variable sets, and their fixed letters are as in Definition 3.1, with this relabeling. For positive numbers $\lambda_1, \dots, \lambda_n$, set $s_J = \sum_{j\in J} \lambda_j$.

Lemma 11.5 (Number of lines)**.** *For a fixed nonempty $J \subset [n]$ there are $3^{n-|J|}$ combinatorial lines with variable set J. The total number of lines in $\mathcal{W}_n$ is $4^n - 3^n$.*

Proof. There are three independent choices for each fixed coordinate. Summing over the nonempty variable sets gives

$$\sum_{J\neq\varnothing} 3^{n-|J|} = \sum_{j=1}^{n} \binom{n}{j} 3^{n-j} = (3+1)^n - 3^n.$$

The variable set and fixed letters are uniquely determined by the line, so this sum counts each line once. □

Lemma 11.6 (A weighted lower bound from density Hales–Jewett)**.** *Let $0 < \delta \le 1$ and choose $n \ge H_3(\delta/4)$ from Theorem 3.19. Let $\lambda_1, \dots, \lambda_n > 0$ and put*

$$S = \max(1, \|M\|) \sum_{j=1}^{n} \lambda_j.$$

Suppose A is a Borel subset of a square Q with $|A| \ge \delta|Q|$, and that $Q' = Q + [-S,S]^2$ satisfies $|Q'| \le 2|Q|$. Then, for $f = \mathbf{1}_A$,

$$\sum_{\varnothing \neq J \subset [n]} 3^{n-|J|} D_{s_J}(f) \ge \frac{\delta}{2}|Q|. \tag{11.1}$$

Proof. Fix θ. Put $d_0 = 0$, $d_1 = u_\theta$, and $d_2 = Mu_\theta$. For a word $w \in \mathcal{W}_n$ let $a_w(\theta) = \sum_j \lambda_j d_{w_j}$, and define the marked fraction

$$m_\theta(x) = 3^{-n} \sum_{w \in \mathcal{W}_n} f(x + a_w(\theta)).$$

Every coordinate of $a_w(\theta)$ has absolute value at most S. Hence $f(x + a_w(\theta))$ is supported in Q' as a function of x. Translation invariance gives

$$0 \le m_\theta \le 1, \qquad \int_{Q'} m_\theta(x)\,\mathrm{d}x = |A| \ge \delta|Q|.$$

Let $G_\theta = \{x \in Q' : m_\theta(x) \ge \delta/4\}$. Splitting the last integral over G_θ and its complement yields

$$\delta|Q| \le |G_\theta| + \frac{\delta}{4}|Q'| \le |G_\theta| + \frac{\delta}{2}|Q|,$$

so $|G_\theta| \ge \delta|Q|/2$.

At each $x \in G_\theta$, at least $(\delta/4)3^n$ words are marked by the condition $f(x + a_w(\theta)) = 1$. Relabeling $\{1,2,3\}$ as $\{0,1,2\}$, Theorem 3.19 therefore supplies a whole marked line. If $\mathcal{L}_n$ is the finite family of all lines, this means

$$\sum_{\ell \in \mathcal{L}_n} \prod_{w \in \ell} f(x + a_w(\theta)) \ge \mathbf{1}_{G_\theta}(x).$$

Integrate over all x; all summands are nonnegative. For a line with variable set J, let $b = \sum_{j \notin J} \lambda_j d_{a_j}$. Its three offsets are b, $b + s_J u_\theta$, and $b + s_J M u_\theta$. The translation $y = x + b$ turns its integral into

$$\int f(y) f(y + s_J u_\theta) f(y + s_J M u_\theta)\,\mathrm{d}y.$$

This expression is independent of the fixed letters. By Lemma 11.5 it occurs $3^{n-|J|}$ times. Integrating next in θ proves (11.1). All functions in this argument are Borel, and all exchanges of finite sums and integrals are justified by nonnegativity or bounded compact support. □

Lemma 11.7 (Selecting scales with close subsums)**.** *Let $\mathcal{S} \subset (0,\infty)$ be unbounded. Given $n \ge 1$ and $\tau > 0$, one can choose $\lambda_1 < \cdots < \lambda_n$ in $\mathcal{S}$ such that, for every nonempty $J \subset [n]$ and $k = \max J$,*

$$\|\Lambda_{\sigma_{s_J} - \sigma_{\lambda_k}}\| < \tau.$$

Proof. By Lemma 11.1, choose $\eta > 0$ so that $1 \le r < 1 + \eta$ implies $\|\Lambda_{\sigma_r - \sigma}\| < \tau$. Choose any $\lambda_1 \in \mathcal{S}$. Having chosen the previous scales, use unboundedness to choose

$$\lambda_k > \max\left(\lambda_{k-1}, \eta^{-1} \sum_{j<k} \lambda_j\right) \quad \text{in } \mathcal{S}.$$

For $k = \max J$ this implies

$$1 \le \frac{s_J}{\lambda_k} \le 1 + \frac{\sum_{j<k} \lambda_j}{\lambda_k} < 1 + \eta.$$

Finally $\sigma_{s_J} - \sigma_{\lambda_k} = (\sigma_{s_J/\lambda_k} - \sigma)_{\lambda_k}$, so the norm equality in Lemma 4.4 gives the desired inequality. When $J = \{1\}$ it is immediate. □

Theorem 11.8 (All sufficiently large planar triangles)**.** *If $E \subset \mathbb{R}^2$ is Lebesgue measurable with $d^*(E) > 0$, then every sufficiently large scale of $\{0, e_1, Me_1\}$ has a translate of a rotation contained in E. Consequently the same conclusion holds for any prescribed noncollinear planar triangle.*

Proof. Fix $0 < \delta < d^*(E)$. Let n be the integer in Lemma 11.6. Suppose for contradiction that the set of missed scales is unbounded. Apply Lemma 11.7 to that set with $\tau = \delta/(4 \cdot 4^n)$ to obtain missed scales $\lambda_1 < \cdots < \lambda_n$.

These scales determine the finite number $S = \max(1, \|M\|) \sum_j \lambda_j$. By Lemma 11.3, choose a square Q of sufficiently large side length that $|E \cap Q| \geq \delta |Q|$ and $|Q + [-S, S]^2| \leq 2|Q|$. For example, the latter condition holds when its side length is at least $2S/(\sqrt{2} - 1)$. Take a Borel set $A \subset E \cap Q$ with $|A| = |E \cap Q|$ and set $f = \mathbf{1}_A$.

For every chosen scale, $D_{\lambda_k}(f) = 0$, by Lemma 11.3. If $J \neq \varnothing$ and $k = \max J$, the endpoint norm bound and the scale selection give

$$0 \leq D_{s_J}(f) = |D_{s_J}(f) - D_{\lambda_k}(f)| \leq \tau \|f\|_2^2 \|f\|_\infty \leq \frac{\delta |Q|}{4 \cdot 4^n}.$$

On summing and using Lemma 11.5,

$$\sum_{J \neq \varnothing} 3^{n-|J|} D_{s_J}(f) \leq \frac{(4^n - 3^n)\delta |Q|}{4 \cdot 4^n} < \frac{\delta}{4} |Q|.$$

This contradicts the lower bound $\delta |Q|/2$ in Lemma 11.6. The missed scales are therefore bounded, proving the first assertion.

For a general triangle, use Lemma 4.2 to write $\Delta = q_0 + \ell U\{0, e_1, Me_1\}$ with $\ell > 0$ and U a rotation. A copy at scale $\lambda\ell$ of the normalized triangle is a translated rotated copy of $\lambda\Delta$, since products and inverses of planar rotations are rotations. Divide the normalized threshold by ℓ. This proves the second assertion and the introductory geometric conclusion. □

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